Meridian Capital

ETL Pipeline Monitoring and Alerting Handbook

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Page 1 -- Executive Overview, Scope, and Governance

Document purpose

This handbook establishes Meridian Capital's standard approach for observing, measuring, and signaling the operational health of ETL pipelines across finance, risk, and enterprise data domains. The content focuses on consistent definitions, shared expectations among support teams, and a traceable record of monitored metrics and alert channels. It is designed to be read alongside the Enterprise Incident Management Standard and the Data Quality Framework Standard. The material reflects routine practices and findings from scheduled reviews and is suitable for general operational reference within the bank.

Parameter set -- Document metadata (Ref: DOC-MCE-DP-HBK-ETL-MON-001)

- Title: ETL Pipeline Monitoring and Alerting Handbook
- Business owner: Enterprise Data Engineering (EDE)
- Technical owner: Data Platform SRE (DPSRE)
- Version: 1.2
- Effective date: 2026-07-15
- Classification: Internal Use Only
- Review cadence: Quarterly
- Next scheduled review: 2026-10-10
- Related records: CHG0039482 (Prod dashboard taxonomy update), PRB0012528 (Q1 schedule adherence analysis), KB-ETL-2147 (Metric definitions)
- Distribution list: DPSRE, EDE, Operations Command Center (OCC), Finance Data Stewardship (FDS), Risk Analytics Technology (RAT), Compliance Data Oversight (CDO)

Executive summary

ETL pipelines at Meridian Capital support daily close processes, regulatory reporting aggregates, liquidity projections, credit risk scoring, and departmental analytics. The monitoring estate spans orchestration health, workload performance, schedule adherence, data quality signals, and business-aligned KPIs used by Finance and Risk. The target state emphasizes early detection of degradation, transparent ownership, and consistent alert semantics. This handbook consolidates naming standards for metrics, dashboards and alert identifiers, and the set of reference severity levels used by OCC for daily interpretation.

Scope and applicability

- In-scope workloads: Batch and micro-batch pipelines orchestrated via standard schedulers used by EDE; data movement and transformation jobs landing in governed storage and warehouse targets; data quality controls instrumented through approved libraries.
- Environments: Development (DEV), Quality Assurance (QA), User Acceptance Testing (UAT), and Production (PROD).
- Out-of-scope: Ad hoc analyst notebooks; one-time backfills performed outside standard orchestration; non-data workloads (e.g., web services) except when they serve as upstream dependencies captured in the pipeline inventory.

Parameter set -- Environment matrix (Ref: ENV-MCE-DP-ETL-001)

- Orchestration: Apache Airflow (managed), standard DAG registry MC-DAG-Prod-01
- Compute: Spark on managed clusters (batch), SQL pushdown on governed warehouses
- Storage: Object storage for staging; governed warehouse zones for curated layers
- Logging: Centralized log index ETL-Core-Prod, retention 180 days
- Metrics: Enterprise monitoring platform (dashboards prefixed DD-Board-ETL-Prod-###)
- Alerting: Notification router integrated with On-Call schedules (PD services PD-SVC-ETL-PROD-01/02)
- Ticketing: ServiceNow (incident INC, change CHG, problem PRB)

Objectives and success criteria

- Reliability: High signal quality for job success/failure, end-to-end latency, and data completeness across daily and intraday cycles.
- Observability: Traceability from alert to pipeline, dataset, owner team, and last change record.
- Consistency: Uniform naming standards for metrics and tags across all managed DAGs and jobs.
- Business alignment: Data freshness and completeness indicators available in time for Finance and Risk cutoffs.
- Cost awareness: Routine review of resource consumption signals to inform capacity planning (non-exhaustive, descriptive).

Stakeholders and roles

- Data Platform SRE (DPSRE): Platform-level health indicators, orchestration metrics, and alert channel stewardship.
- Enterprise Data Engineering (EDE): Job-level monitoring, data quality rule registration, pipeline ownership in inventory.
- Operations Command Center (OCC): Daily health reviews, first-line interpretation of alerts, status consolidations.
- Finance Data Stewardship (FDS): Business validation signals for close activities; source-of-truth attestations.
- Risk Analytics Technology (RAT): Risk model feed completeness checks and cut-off windows.
- Compliance Data Oversight (CDO): Monitoring of controls related to regulatory datasets (descriptive only).

Parameter set -- Primary monitoring channels (Prod snapshot, 2026-07-19T06:30:00Z)

- Dashboard: DD-Board-ETL-Prod-042 (Daily Batch Health)
- Dashboard: DD-Board-ETL-Prod-057 (Data Freshness -- Finance)
- Dashboard: DD-Board-ETL-Prod-063 (Risk Aggregations Window)
- Alert service: PD-SVC-ETL-PROD-01 (Batch Core), PD-SVC-ETL-PROD-02 (DQuality)
- Log view: ETL-Core-Prod -- source: airflow_task, spark_job
- Ticket queue: SNOW-ETL-OPS (group), OCC-ETL-DAILY (assignment)

Systems and tooling overview

Monitoring for ETL pipelines relies on a standardized tag model (application, pipeline_id, dataset, domain, env, owner) to unify signals across orchestration, compute, and quality layers. Dashboards aggregate runtime statistics (start lag, duration distributions, success ratio, retry counts), while alert rules track adherence to committed windows. OCC performs daily sign-offs using a consolidated view of pipelines tagged as Finance Close critical, and EDE maintains the mapping of pipelines to business processes in the inventory repository. All changes to alert thresholds follow the bank's change management process and are reflected in the dashboard annotations for auditability.

Parameter set -- Reference inventory extract (non-exhaustive)

- Pipeline ID: ETL_FIN_GL_DAILY, Owner: EDE-Finance, Domain: Finance, Frequency: Daily, Criticality: Close-Critical
- Pipeline ID: ETL_RISK_PD_INTRADAY, Owner: RAT-ETL, Domain: Risk, Frequency: Hourly, Criticality: Regulatory-Support
- Pipeline ID: ETL_TREASURY_LIQ_WEEKLY, Owner: EDE-Treasury, Domain: Treasury, Frequency: Weekly, Criticality: High

Governance and review cadence

- Quarterly monitoring review: Confirms metric definitions, dashboard coverage, and severity alignment. Last held 2026-07-11 (Minutes: MRN-ETL-MON-QR-2026Q3; Action Tracker: AT-QR-2026Q3-ETL).
- Weekly OCC summary: Aggregates alert counts, mean time-to-acknowledgment indicators, and notes on recurring patterns. Reference log: OCC-SUM-ETL-2026W28.
- Change record linkage: Dashboard and alert updates are associated with CHG records; notable recent updates include CHG0039482 (severity alignment), CHG0039527 (data freshness thresholds for Finance Close).

Step catalog (descriptive references; see Appendix B for full list)

- Step ID: MCE-ETL-MON-S01 -- Description: Pipeline inventory curation and tag baseline confirmation. Outcome: Up-to-date owner and criticality mapping.
- Step ID: MCE-ETL-MON-S02 -- Description: Metric coverage validation against the core set (start lag, duration, success ratio, freshness, row deltas). Outcome: Coverage report in knowledge base.
- Step ID: MCE-ETL-MON-S03 -- Description: Alert taxonomy conformity check. Outcome: Severity labels consistent with OCC dashboard filters.
- Step ID: MCE-ETL-MON-S04 -- Description: Review of annotation history for threshold changes. Outcome: Traceable linkage to approved changes.

Recent review notes (extract)

- 2026-07-12T14:05:22Z [DPSRE]: Batch runtime percentile bands updated for ETL_FIN_GL_DAILY; historical baselines extended to 180 days (PRB0012528).
- 2026-07-13T09:32:10Z [EDE-Finance]: Added data completeness metric for subledger feed; tag dataset=subledger_v3 (CHG0039563).
- 2026-07-13T11:47:55Z [OCC]: Severity label for freshness drift on DD-Board-ETL-Prod-057 now P2 for Close-Critical datasets (KB-ETL-2147 alignment).

Page 2 -- Pipeline Landscape, Identifiers, and Alert Taxonomy

Pipeline landscape and identifiers

Meridian Capital maintains a canonical naming convention to simplify monitoring and reporting. Each pipeline identifier encodes business domain, dataset, and frequency, ensuring consistent correlation across metrics, logs, and tickets. The convention is descriptive and minimizes ambiguity during cross-team reviews.

Parameter set -- Identifier schema

- Format: ETL_{DOMAIN}_{DATASET}_{FREQUENCY}
- Domains (examples): FIN, RISK, TRSY, RET
- Datasets (examples): GL, PD, LCR, CARDS
- Frequencies: INTRADAY, HOURLY, DAILY, WEEKLY, MONTHLY
- Tag set (minimum): application=etl, pipeline_id, dataset, domain, env, owner, criticality

Primary workflow categories

- Ingestion: Source extracts from governed upstreams into staging zones using standardized connectors and manifests.
- Transformation: Batch transformations executed on managed compute or warehouse pushdown with lineage annotations.
- Load: Materialization into curated layers and downstream marts, with row-count and checksum signals emitted.
- Data quality: Rule evaluations for freshness, completeness, and referential consistency; results surfaced as metrics and dashboard tiles.
- Business validation: Finance and Risk steward attestations captured as status flags aligned to cutoff calendars.

Metrics dictionary (extract; see Appendix C for comprehensive list)

- job_start_lag_seconds: Time difference between scheduled start and actual start of a job within a pipeline.
- job_duration_seconds: Elapsed processing time for a job execution.
- pipeline_success_ratio: Ratio of successful jobs to total jobs in the last defined window.
- data_freshness_lag_seconds: Age of the most recent successful load compared to expected data timestamp.
- row_count_delta_ratio: Proportional change in loaded row counts relative to a rolling baseline.
- dq_rule_pass_ratio: Percentage of data quality rules passing in the last run.
- retry_count: Number of task retries observed for the current execution window.

SLO catalog excerpt (Prod, 2026-07 snapshot)

- ETL_FIN_GL_DAILY: Completion by 05:30 UTC on business days; freshness lag under 60 minutes; dq_rule_pass_ratio >= 0.98; monthly target: 99% adherence.
- ETL_RISK_PD_INTRADAY: Windowed delivery every hour between 06:00-20:00 UTC; end-to-end latency under 20 minutes; dq_rule_pass_ratio >= 0.99; monthly target: 99.5% adherence.
- ETL_TRSY_LIQ_WEEKLY: Completion by Monday 12:00 UTC; data completeness flagged for all inputs; dq_rule_pass_ratio >= 0.97; monthly target: 99% adherence.

Parameter set -- Alert taxonomy (KB-ETL-2147-aligned)

- Severity P1: Material impact to regulatory submission timing or finance close; immediate OCC visibility;

PD-SVC-ETL-PROD-01.

- Severity P2: High priority degradation risking SLO breach for critical datasets; OCC summary flag on DD-Board-ETL-Prod-042.
- Severity P3: Non-critical schedule drift or transient freshness variance; tracked in weekly OCC summary.
- Severity P4: Informational deviations and post-change annotations; consolidated in monthly review.

Alert rule catalog (summary extract; platform-registered)

- Rule ID: AR-ETL-0421 -- Signal: data_freshness_lag_seconds -- Scope: ETL_FIN_GL_DAILY -- Threshold: >3600s for Close-Critical -- Cooldown: 30m -- Owner: EDE-Finance
- Rule ID: AR-ETL-0578 -- Signal: pipeline_success_ratio -- Scope: domain=RISK -- Threshold: <0.98 over last 6h -- Cooldown: 15m -- Owner: RAT-ETL
- Rule ID: AR-ETL-0632 -- Signal: job_start_lag_seconds -- Scope: env=PROD, frequency=DAILY -- Threshold: >900s -- Cooldown: 20m -- Owner: DPSRE
- Rule ID: AR-ETL-0714 -- Signal: dq_rule_pass_ratio -- Scope: dataset=subledger_v3 -- Threshold: <0.98 -- Cooldown: 60m -- Owner: EDE-Finance

Parameter set -- Dashboard inventory (Prod, selected)

- DD-Board-ETL-Prod-042: Daily Batch Health -- sections: Start Lag, Runtime P95, Success Ratio Trends, Retry Heatmap
- DD-Board-ETL-Prod-057: Data Freshness -- Finance -- sections: Freshness Lag by Dataset, Cutoff Alignment, Annotation Stream
- DD-Board-ETL-Prod-063: Risk Aggregations -- sections: Intraday Delivery Window, Completeness, Rolling Variance
- DD-Board-ETL-Prod-075: Quality Overview -- sections: Rule Coverage, Failure Topology, Owner Mapping

Daily health review log (OCC extract, 2026-07-19)

- 2026-07-19T05:12:08Z [OCC-Dashboard] DD-Board-ETL-Prod-042 reports 98.7% pipeline_success_ratio rolling 24h; retries elevated for ETL_FIN_GL_DAILY (post-change CHG0039482 annotation present).
- 2026-07-19T05:28:44Z [OCC-AlertRouter] AR-ETL-0632 triggered for two non-critical daily jobs; classified P3 due to non-Critical tag; consolidated into weekly summary ticket TASK-ETL-2026W29-01.
- 2026-07-19T05:49:21Z [OCC-Quality] dq_rule_pass_ratio at 0.985 for Finance datasets; subledger_v3 freshness within target (AR-ETL-0714 not triggered).
- 2026-07-19T06:02:33Z [OCC-Summary] Close-Critical pipelines within schedule window; no P1/P2 events recorded for the period 00:00-06:00 UTC; reference report OCC-SUM-ETL-2026W29-D1.

Dependencies and upstream notes

- Source system calendar alignment is reflected via business day parameters; monitoring tiles annotate non-business days and adjusted cutoffs.
- For intraday risk feeds, upstream market data windows are tagged as external dependencies to contextualize freshness signals in DD-Board-ETL-Prod-063.
- Baseline models for runtime and row deltas incorporate the last 90-180 days depending on pipeline volatility; annotations list the effective window length for each metric.

Step catalog (descriptive references)

- Step ID: MCE-ETL-MON-S05 -- Description: Calibration of seasonality-aware runtime bands for daily and

month-end profiles. Outcome: Updated dashboard bands and consistent alert sensitivity across cycles.

- Step ID: MCE-ETL-MON-S06 -- Description: Review of pipeline-to-business-process mapping for Close-Critical designation accuracy. Outcome: Adjusted criticality tags with steward acknowledgment.
- Step ID: MCE-ETL-MON-S07 -- Description: Verification of logging tag completeness for new or modified DAGs. Outcome: Confirmed metric joinability and inventory linkage.

Findings from recent reviews (summary)

- Batch variability: Runtime P95 stabilized after configuration normalization; OCC notes reduced alert volume associated with start lag deviations.
- Freshness drift: Finance datasets met the 60-minute freshness target during the last five business days; annotation trail indicates proactive cutover planning for month-end.
- Quality coverage: Rule coverage sustained above 95% for Close-Critical datasets; two informational items logged regarding emerging dimensions in subledger_v3 (tracked via TASK-ETL-2026W29-05).

Parameter set -- Ownership and contact (reference)

- PD service: PD-SVC-ETL-PROD-01 (Batch Core) -- Primary: DPSRE -- Secondary: EDE-Finance
- PD service: PD-SVC-ETL-PROD-02 (DQuality) -- Primary: EDE -- Secondary: DPSRE
- Ticket group: OCC-ETL-DAILY -- Purpose: Daily consolidation and summary records
- Knowledge bases: KB-ETL-2147 (Taxonomy), KB-ETL-2193 (Dashboard usage conventions), KB-ETL-2201 (Metric dictionary)

Notes on compliance and auditability

Monitoring definitions, severity labels, and inventory mappings are archived in linked knowledge articles and are updated through standard change records. Dashboards display annotation streams aligned to approved changes, providing traceability for audit queries. The OCC daily health review retains summary logs and rollups used in management reporting. All references in this handbook are descriptive and align with the bank's routine operational practices.

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Document Section: Monitoring Architecture and Signal Taxonomy

Prepared by: Data Engineering -- ETL Operations, with SRE Platforms

Last Reviewed: 2026-07-17 09:32 UTC (Ref: MC-DATA-REV-2417)

2. Monitoring Architecture Overview

This section summarizes the end-to-end monitoring surfaces that provide operational visibility into the Enterprise ETL Platform (Program ID: ETL-PF-01). The objective is to ensure consistent observability across source ingestion, transformation orchestration, quality controls, and downstream publishing. The ETL Platform is treated as a business-critical service, with monitoring aligned to service-level objectives (SLOs) and capacity/volume baselines. The architecture emphasizes signal layering: component health (infrastructure, orchestration), pipeline runtime (task/job metrics), data contract adherence (schema, lineage), and business-facing data reliability (freshness, completeness, accuracy).

All monitors are cataloged within the Monitoring Registry (REG-MON-ETL v2.3). Changes to monitors, alert thresholds, or routing rules are governed via standardized change tickets (CHG-ETL- series) and require approval from the ETL Ops Duty Manager and SRE On-Call.

Component Inventory (Monitoring Scope)

- Scope includes ingestion connectors, batch schedulers, transformation jobs, quality gates, and publishing endpoints.
- Component criticality ratings are aligned to the Revenue Risk Index (RRI) and Regulatory Reporting importance.

Component Inventory Table -- ETL-PF-01

- Component ID: SRC-FX-ING-01

Owner: Data Integration Team (DI-NA)

Criticality: High

Uptime SLO: 99.9% monthly

Primary Monitors: Availability, Throughput, Backlog Depth

Monitor IDs (Registry): MON-AV-1203; MON-TH-2210; MON-BL-3308

- Component ID: ORC-BAT-CTL-02

Owner: ETL Operations (ETLOPS)

Criticality: High

Uptime SLO: 99.95% monthly

Primary Monitors: Scheduler Heartbeat, Queue Latency, Task Failure Rate

Monitor IDs (Registry): MON-HB-1407; MON-QL-2712; MON-TF-4419

- Component ID: TRF-CRD-RSK-07

Owner: Credit Risk Data Engineering (CRDE)

Criticality: High

Uptime SLO: 99.9% monthly

Primary Monitors: Job Duration, Row-Level Error Rate, Checkpoint Success

Monitor IDs (Registry): MON-DU-3516; MON-ER-3619; MON-CK-3721

- Component ID: DQ-GATE-CUR-01

Owner: Data Quality Office (DQO)

Criticality: High

Uptime SLO: 99.9% monthly

Primary Monitors: Data Completeness Score, Dimension Conformance, Anomaly Flags

Monitor IDs (Registry): MON-DC-4102; MON-DM-4105; MON-AN-4111

- Component ID: PBL-MART-LQ-09

Owner: Analytics Publishing (APUB)

Criticality: Medium

Uptime SLO: 99.5% monthly

Primary Monitors: Publish Latency, Consumer Readiness, Version Consistency

Monitor IDs (Registry): MON-PL-5207; MON-CR-5210; MON-VC-5218

Signal Taxonomy

Signals are grouped to ensure consistent thresholding and incident routing. Each signal category includes standard labels: component_id, environment (prod/non-prod), dataset, owner_team, and severity_candidate.

- Availability: Binary liveness and health checks for schedulers, agents, and endpoints.
- Latency: End-to-end and segment durations relative to SLOs; includes schedule delay and run time.
- Volume: Record counts, file counts, and bytes processed; measured against rolling baselines.
- Quality: Completeness, uniqueness, validity against data contracts; summarized via percentile scores.
- Schema Drift: Additions, removals, or type changes in inbound structures; contract variance is flagged.
- Backlog: Queue depth and unprocessed work units relative to time-based drains.
- Freshness: Age of latest successful partition/publication versus consumer expectations.

Signal Mapping Table -- Monitoring Registry (Excerpt)

- Signal: Availability

Metric Source: Orchestration Heartbeat; Health Probe

Threshold Boundary: < 1 missed heartbeat within 5 min

Sampling Window: 5 min rolling

Escalation Group: SRE Primary (SRE-ETL-OC)

- Signal: Latency

Metric Source: Job Start/End Markers; Schedule Delay

Threshold Boundary: p95 runtime <= baseline + 20%

Sampling Window: 24 hr rolling, hourly refresh

Escalation Group: ETL Operations (ETLOPS)

- Signal: Volume

Metric Source: Ingestion Counts; Byte Totals

Threshold Boundary: -15% to +30% versus 14-day median

Sampling Window: Daily, with intraday checkpoints

Escalation Group: Data Integration (DI-NA)

- Signal: Quality

Metric Source: DQ Assertions; Rule Scores

Threshold Boundary: Composite DQ Score >= 98.5

Sampling Window: Per-batch evaluation

Escalation Group: Data Quality Office (DQO)

- Signal: Schema Drift

Metric Source: Contract Validator

Threshold Boundary: Zero unapproved breaking changes

Sampling Window: Per-ingest event

Escalation Group: Data Contract Stewards (DCS)

KPI and SLO Definitions

Service reliability is measured via KPIs anchored to business outcomes. The following KPIs inform the alerting thresholds and incident severities. All SLOs are set for the production environment (ENV: PROD-NA-1) and reviewed quarterly.

KPI/SLO Table -- ETL-PF-01

- KPI: Daily Ingestion Freshness (FX and Equities)

Definition: Max age (minutes) between market close and curated availability

Target SLO: <= 75 minutes p95; <= 110 minutes p99

Data Source: Freshness Markers (FRESH-MON-201)

Owner: ETLOPS

- KPI: Transformation Success Rate

Definition: Percentage of completed jobs without failure retry

Target SLO: >= 99.7% daily

Data Source: Orchestration Audit (AUD-ORC-502)

Owner: CRDE

- KPI: Data Completeness Score

Definition: Weighted completeness across critical tables (core fact + reference)

Target SLO: >= 99.0 daily composite; no critical table < 98.0

Data Source: DQ Rule Engine (DQ-ENG-310)

Owner: DQO

- KPI: Consumer Publish Latency

Definition: Time from final transform completion to consumer-ready publish

Target SLO: <= 20 minutes p95

Data Source: Publish Tracker (PUB-TRK-119)

Owner: APUB

Dashboard Catalog (Operational)

Dashboards are versioned assets within the Analytics Monitoring Portal. URLs are cataloged internally under DASH-ID; access managed by role.

Dashboard Summary Table

- DASH-ID: DB-ETL-OVR-01

Purpose: Executive overview -- SLO attainment, incident trend, backlog status

Refresh Cadence: 5 min

Primary Audience: Data Leadership, Incident Command

- DASH-ID: DB-ING-CHN-04

Purpose: Source ingestion channels -- throughput, error rate, schema drift

Refresh Cadence: 1 min

Primary Audience: Data Integration

- DASH-ID: DB-ORC-RUN-02

Purpose: Orchestration timelines -- schedule delay, concurrency, task failures

Refresh Cadence: 1 min

Primary Audience: ETLOPS, SRE

- DASH-ID: DB-DQ-SCO-03

Purpose: DQ composite scores and rule-level drilldowns

Refresh Cadence: Per batch

Primary Audience: DQO, Data Stewards

- DASH-ID: DB-PUB-LAT-05

Purpose: Publish pipeline -- release versions, consumer readiness

Refresh Cadence: 2 min

Primary Audience: APUB

Notes and References

- Change Log: CHG-ETL-11872 (Added p95 transformation latency lens, 2026-06-29)
- Incident Learning: MC-DATA-INC-2026 (Backlog saturation misclassified; threshold tuned 2026-07-06)
- Ownership Roster: OWN-ETL-2026Q3 (Refreshed team contacts and escalation paths)

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Document Section: Alert Routing, Severity Model, and Operational States

Prepared by: SRE Platforms with ETL Operations

Last Reviewed: 2026-07-18 14:05 UTC (Ref: MC-SRE-REV-3309)

3. Alert Routing and Severity Model

Alerting adheres to a standardized severity model calibrated to customer impact, data timeliness, and regulatory considerations. Severity levels determine notification channels, response time objectives, and status reporting cadence. All production alerts must reference a Monitor ID and include required labels (component_id, dataset, environment, owner_team, severity_candidate, region).

Severity Matrix

- SEV-1 (Critical): Material impact on regulatory or revenue-sensitive datasets; missed freshness SLO p99; complete pipeline halt without workaround.

Response Objective: Engage Incident Command within 5 minutes; executive notification required.

Routing: SRE Primary + ETLOPS + Data Leadership bridge.

Ticket: Auto-create Major Incident (MI) -- prefix MC-DATA-MI-.

- SEV-2 (High): Degradation impacting SLO p95; partial pipeline failure; repeat retries exceeding budget; consumer publish latency beyond threshold for priority marts.

Response Objective: Acknowledge within 10 minutes; recovery actions prioritized.

Routing: ETLOPS On-Call + owning data domain.

Ticket: Incident (INC) -- prefix MC-DATA-INC-.

- SEV-3 (Moderate): Isolated task failures with successful retries; small variance from volume baselines; non-critical data set delays.

Response Objective: Acknowledge within 30 minutes; monitor and mitigate.

Routing: Owning team; informational to SRE.

Ticket: Service Request or Problem (SR/PRB) upon recurrence.

- SEV-4 (Low/Info): Transient anomalies, known maintenance windows, expected seasonal effects.

Response Objective: No immediate action; catalog for trend analysis.

Routing: Owner only.

Ticket: Optional knowledge entry (KB-DATA-).

Alert Rule Catalog (Excerpt)

- Rule Name: ETL-Orchestrator-Heartbeat

Monitor ID: MON-HB-1407

Condition: Two consecutive missed heartbeats within 5-minute window

Default Severity: SEV-1

Notification Channels: Pager (SRE), Collaboration Bridge, Email Digest

Ticket Template: MC-DATA-MI-<auto>

- Rule Name: Freshness-Breach-Core-FX

Monitor ID: MON-FR-2009

Condition: p95 freshness > 75 min for FX curated tables

Default Severity: SEV-2 (escalate to SEV-1 if p99 > 110 min)

Notification Channels: Pager (ETLOPS), Channel Alert

Ticket Template: MC-DATA-INC-<auto>

- Rule Name: DQ-Composite-Dip

Monitor ID: MON-DC-4102

Condition: Composite DQ Score < 98.5; any critical table < 98.0

Default Severity: SEV-2

Notification Channels: DQO Group, Email Digest

Ticket Template: MC-DATA-INC-<auto>

- Rule Name: Volume-Deviation-Moderate

Monitor ID: MON-TH-2210

Condition: -15% to -25% vs 14-day median for ingestion channel

Default Severity: SEV-3

Notification Channels: Owner Team notice

Ticket Template: SR-DATA-<auto>

Alert Lifecycle States

Alerts progress through standardized operational states for accountability and auditability. All state transitions are captured in the Monitoring Registry and mirrored in the incident ticket.

Lifecycle State Definitions

- Firing: Signal exceeds defined threshold; alert created with contextual labels and recent metric snapshots.
- Acknowledged: An engineer has accepted responsibility; timer shifts from "time to page" to "time to mitigate."
- Mitigated: Underlying signal returned to within threshold; post-mitigation observation begins.
- Resolved: Incident ticket closed following validation checklist; residual risk logged if applicable.
- Verified: Post-resolution validation by owner or DQO confirms data consumer readiness.

Triage Worksheet (Operational Steps)

- Step ID: TR-01

Action: Confirm component health via dashboard DB-ORC-RUN-02 (scheduler status, queue latency)

Tool: Analytics Monitoring Portal

Expected Signal: Handshake timestamp updated within last 60 seconds

Max Elapsed: 5 minutes

Ticket Note: Paste dashboard snapshot ID

- Step ID: TR-02

Action: Validate scope of impact across datasets listed in component labels

Tool: Dashboard DB-ETL-OVR-01

Expected Signal: Single-domain impact or cross-domain markers

Max Elapsed: 10 minutes

Ticket Note: Record impacted dataset names and approximate windows

- Step ID: TR-03

Action: Assess data quality posture using DB-DQ-SCO-03

Tool: DQ dashboard

Expected Signal: Rule failures correlated with impacted runs

Max Elapsed: 10 minutes

Ticket Note: Reference failing rule IDs

- Step ID: TR-04

Action: Check publish readiness indicators on DB-PUB-LAT-05

Tool: Publishing dashboard

Expected Signal: Consumer-ready flag state; version drift status

Max Elapsed: 5 minutes

Ticket Note: Note publish version and latency percentile

- Step ID: TR-05

Action: Determine severity candidate and confirm routing

Tool: Incident Management System

Expected Signal: Severity aligned to matrix; proper groups notified

Max Elapsed: 5 minutes

Ticket Note: Document reasoning for severity selection

Known Benign Anomaly Patterns

The following patterns are cataloged to reduce false positives while preserving sensitivity to material risks. Suppression or threshold adjustments must proceed via change control (CHG-ETL- series) and include a rollback plan.

Benign Patterns Table

- Pattern: End-of-month volume surge in credit risk feeds (TRF-CRD-RSK-07)

Typical Window: Last business day 20:00-23:00 UTC

Expected Effect: +25% to +40% volume; p95 runtime +15%

Handling: Pre-approved SEV-3 downtier if within documented band (CHG-ETL-11791)

- Pattern: Market holiday reduced activity (FX, Equities)

Typical Window: Published holiday calendar days

Expected Effect: -30% to -60% volume; freshness unchanged

Handling: Suppression window per calendar (CHG-ETL-11803)

- Pattern: Reference data rebuild (PBL-MART-LQ-09)

Typical Window: Weekly 01:00-02:00 UTC

Expected Effect: Temporary version drift; publish latency +10 min

Handling: Informational alerts only (SEV-4), with consumer notice

Maintenance Windows and Alert Suppression

Maintenance windows are documented with explicit component scope, time bounds, and owner approvals. Suppression rules are time-bound and auditable.

Maintenance/Suppression Table

- Window ID: MW-ETL-2026-07-21-A

Components: ORC-BAT-CTL-02

Time (UTC): 01:00-01:30

Approved By: ETLOPS Duty Manager

Suppression: Heartbeat, Queue Latency (read-only monitors remain active)

Reference: CHG-ETL-11912

- Window ID: MW-ETL-2026-07-28-B

Components: DQ-GATE-CUR-01

Time (UTC): 02:00-02:20

Approved By: DQO Lead

Suppression: DQ Composite; Rule Drilldowns

Reference: CHG-ETL-11944

On-Call and Escalation Summary (Q3)

- Primary: SRE-ETL-OC (24x7 paging)
- Secondary: ETLOPS (follow-the-sun rotation)
- Domain Owners: DI-NA (ingestion), CRDE (credit risk), DQO (quality), APUB (publishing)
- Duty Manager Roster: OWN-ETL-2026Q3 (see roster for contact matrix)
- Executive Notification Path: Data Leadership bridge (SEV-1 only), status cadence every 15 minutes until Mitigated

Recent Events and Adjustments

- 2026-07-06 03:14 UTC -- MC-DATA-INC-2026: Backlog saturation alert tuned for FX ingestion based on seasonal trend analysis; volume deviation bands updated to 15/30% (Ref: CHG-ETL-11872).
- 2026-07-12 22:47 UTC -- MC-DATA-PRB-1310: Schema drift false positive on non-breaking field addition; contract validator updated to honor optional fields list (Ref: CHG-ETL-11889).

All alert rule changes require regression validation against the prior 30-day signal history. Results are appended to the change ticket and cross-referenced in the Monitoring Registry for audit.

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Page 5: Data Freshness and Latency Monitors

Overview

Data freshness and latency monitors verify that pipeline outputs arrive within agreed service windows and reflect recent source activity. This section defines canonical metrics, environment parameters, monitoring thresholds, and the routine review cadence adopted by Data Platform Operations (DPO) and Enterprise Data Engineering (EDE). The objective is to consistently detect delayed or stale datasets ahead of downstream BI refreshes and intraday risk computations.

Key Definitions

- Freshness: Current UTC time minus the maximum available business event timestamp (watermark) in the target dataset.
- Latency: Duration between the scheduled pipeline start and the time the last committed record for the target partition is available in the warehouse.
- SLA Window: The time interval during which the dataset must be available for dependent consumers.

Environment Parameters

Parameter	Value	Notes
Environment	PROD, UAT	Monitors are active in both; PROD alerts a..
Time Standard	UTC	All timestamps are recorded in UTC.
Business Calendar	MC-Global-2026	Holidays affect threshold relaxations (non-..
Monitoring Platform	Meridian Observability (MO) v3.2	Metric store and alerting rules engine.
Data Warehouse	MC Lakehouse Cluster: DW-PROD-01	Primary analytical store for monitored targ..
Ticketing System	ServiceCenter Cloud	Incident prefix INC-, Problem prefix PRB-, ..

Canonical Metrics and Identifiers

Metric ID	Description	Unit	Collection Frequency
m.freshness_minutes	Current time minus dataset wa..	minutes	Every 5 minutes
m.latency_minutes	Last partition availability minus..	minutes	Every 5 minutes
m.watermark_timestamp	Max event_time observed in ta..	ISO8601	Every 5 minutes
m.ingest_lag	Source landing time minus extr..	minutes	Every 5 minutes

Monitored Pipelines and Thresholds

Pipeline ID	Dataset	SLA Window (UTC)	Freshness Threshold	Latency Threshold	Owners
ETL-FX-01	Trades_FX_Daily	06:00-06:30	<= 90 minutes	<= 30 minutes	EDE-FX, DPO-East
ETL-EQ-03	Trades_Equities_T+0	06:15-06:45	<= 75 minutes	<= 35 minutes	EDE-Equities, DPO
ETL-CUST-07	Custody_Balance_Sn..	05:45-06:15	<= 120 minutes	<= 25 minutes	EDE-Custody, DPO
ETL-REF-02	Reference_Security_..	04:30-05:00	<= 240 minutes	<= 20 minutes	EDE-RefData, DPO
ETL-IR-05	Rates_Curve_Intraday..	Top of hour+10	<= 65 minutes	<= 10 minutes	EDE-Rates, DPO-West

Alert Routing Matrix (Freshness/Latency)

Severity	Condition Example	Notification Path	Ticket Auto-Create
Sev-2	SLA breach > 2x threshold (P..	DPO On-Call, EDE Owner, BI ..	INC
Sev-3	SLA breach <= 2x threshold or..	DPO On-Call, EDE Owner	INC
Sev-4	UAT breach or PROD warning ..	DPO Slack Channel #dpo-alerts	Task (TSK)

Daily Review Procedure (Freshness/Latency)

Step	Action Summary	Actor	Tooling/Artifact	Expected Outcome	Target Duration
1	Validate prior-day SLA..	DPO Duty Analyst	MO Dashboard: Fresh..	Conformance summar..	10 min
2	Cross-check latency v..	DPO Duty Analyst	MO Jobs Timeline, Sc..	Confirm late start vs lo..	8 min
3	Annotate non-trading ..	DPO Duty Analyst	Business Calendar, R..	Mark accepted deviati..	5 min
4	Escalate material devi..	DPO On-Call	ServiceCenter Ticket li..	Open/updated INC wit..	5 min
5	Notify stakeholders for..	DPO On-Call	Distribution List: BI/Re..	BI aware of potential d..	3 min

Sample Alert Digest (Past 24h)

Alert ID	Pipeline	Triggered At (UTC)	Metric	Observed	Threshold	Status	Ticket
A-20260712-1	ETL-FX-01	2026-07-12T06:21:..	m.freshness_minut..	104	90	Resolved	INC-44521
A-20260712-2	ETL-IR-05	2026-07-12T11:10:..	m.latency_minutes	12	10	Acknowledged	INC-44524
A-20260712-3	ETL-REF-02	2026-07-12T04:48:..	m.watermark_time..	2026-07-11T23:00Z	>= 2026-07-12T00..	Informational	TSK-98211

Event Timeline Example (ETL-FX-01, 2026-07-12)

Timestamp (UTC)	Event	Source
06:00:00	Scheduled start	Scheduler Run MC-SCH-01
06:02:33	Extract started	Job Log JL-593844
06:14:05	Transform completed	Job Log JL-593845
06:18:49	Load completed (partition d=2026-07-11)	Warehouse Audit WA-7772
06:21:12	Freshness alert fired (104 min > 90 min)	MO Alert A-20260712-1
06:27:40	Annotation: Late vendor file by 12 minutes	INC-44521
06:33:19	Alert auto-resolved after data reconciliation	MO Alert A-20260712-1

Findings and Notes (Week of 2026-07-06)

- ETL-FX-01 experienced two marginal freshness overruns linked to vendor cutover testing. No downstream BI refresh missed the reporting window. PRB-1321 opened to track vendor schedule stability. Temporary 10-minute soft window applied for non-trading Mondays per MC-Global-2026.
- ETL-IR-05 hourly job intermittently exceeds latency by 1-2 minutes during top-of-hour contention with equities intraday backfills. Capacity advisory logged (CAP-771) with Platform Compute to evaluate queue alignment for 11:00-15:00 UTC.
- ETL-REF-02 maintained compliance; watermark timestamp remained within expected overnight sync parameters even during reference enrichment runs.

Governance and Metrics Retention

- Freshness and latency time series retained at 5-minute granularity for 180 days in MO.
- SLA evaluation artifacts (conformance summaries, exceptions, stakeholder notifications) retained with associated tickets for 24 months for auditability.
- Labels applied to all alerts: pipeline_id, dataset, environment, service_tier, business_domain.

Quarterly KPI Targets (Freshness/Latency)

KPI	Target (Q3-2026)	Responsible Team
SLA conformance (Tier-1 datasets)	>= 99.0%	DPO + EDE
Mean time to acknowledge (Sev-2/3)	<= 5 minutes	DPO On-Call
Mean time to resolve (Sev-3)	<= 45 minutes	DPO/EDE Joint
False-positive rate (freshness)	<= 2.0%	Observability Eng.

Escalation Contacts

Team	Primary Contact (UTC)	Coverage Window	Channel
Data Platform Ops	DPO-OnCall-Group	24x5	Pager, #dpo-incident-prod
Enterprise Data Eng.	EDE-Domain-OnCall (per dom..	06:00-22:00	Pager, #ede-runtime
BI and Reporting	BI-Duty-Manager	06:00-18:00	Email DL: BI-Alerts

All deviations and accepted exceptions are documented against the relevant ticket with timestamps, rationale, and reviewer name. The DPO Duty Analyst ensures that business calendar effects are consistently annotated to maintain reliable SLA reporting.

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Page 6: Volume and Schema Drift Monitoring

Overview

Volume and schema drift monitors ensure that data completeness and structural consistency remain within expected bounds for each dataset and ingest feed. Volume monitors detect anomalous record counts, partition sparsity, or distributional changes. Schema drift monitors detect column additions/removals, data type changes, and primary key constraint violations. This section covers baseline calculations, anomaly parameters, review cadence, and the decision matrix for triage.

Signal Definitions

- Volume Delta (%): Relative difference between observed record_count and expected baseline for the same business date and comparable weekday/season.
- Null Ratio (%): Proportion of nulls in critical fields compared to baseline tolerance.
- Duplicate Rate (%): Fraction of records breaching declared uniqueness constraints.
- Schema Change: Detected modification to column set or types versus the catalog contract at runtime.
- Key Violation: Existence of duplicate business keys in the output partition.

Anomaly Engine Parameters

Parameter	Default (Tier-1)	Default (Tier-2)	Notes
Baseline Window	28 days rolling	21 days rolling	Seasonality-aware (weekday w..
Sensitivity (Z-score)	2.5	3.0	Lower value = higher sensitivity
Minimum Counts (records)	10,000	2,000	Avoids spurious alerts on small..
Null Ratio Tolerance	+2.0 p.p.	+3.5 p.p.	Absolute percentage point delt..
Duplicate Rate Tolerance	+0.25 p.p.	+0.50 p.p.	Applies to business keys
Suppression Window (cooldow..	45 minutes	30 minutes	Prevents alert storms for the s..
Change Ticket Linkage Requir..	Yes	Yes	Schema changes require prior ..

Monitored Datasets and Baselines

Dataset	Pipeline	Expected Count (We..	Expected Count (We..	Critical Fields	Tier
Trades_FX_Daily	ETL-FX-01	1.2M +/- 10%	0.2M +/- 15%	trade_id, ccy_pair	1
Trades_Equities_T+0	ETL-EQ-03	5.4M +/- 8%	0.8M +/- 12%	trade_id, isin	1
Custody_Balance_Sn..	ETL-CUST-07	40M +/- 5%	40M +/- 5%	account_id, as_of_date	1
Reference_Security_..	ETL-REF-02	1.8M +/- 3%	1.8M +/- 3%	sec_id, asset_class	2
Rates_Curve_Intraday..	ETL-IR-05	35k +/- 12% (per hour)	25k +/- 12% (per hour)	curve_id, tenor	1

Recent Alerts (Volume/Schema) with Outcomes

Alert ID	Dataset	Triggered At (UTC)	Signal	Observed	Baseline/Thresh.	Status	Ticket Outcome S..
A-20260711-9	Trades_Equities_T..	2026-07-11T16:49:..	Volume Delta (%)	-12.8%	+/-8% (Tier-1)	Resolved	INC-44498 Vendor..
A-20260712-5	Custody_Balance_..	2026-07-12T06:03:..	Duplicate Rate (%)	0.36%	+0.25 p.p.	Investigate	INC-44527 Pendin..
A-20260712-7	Reference_Securit..	2026-07-12T04:41:..	Schema Change	+1 column	CHG-98322 expec..	Informational	TSK-98256 Chang..
A-20260712-8	Rates_Curve_Intra..	2026-07-12T12:12:..	Null Ratio (%)	+2.4 p.p.	+2.0 p.p.	Acknowledged	INC-44530 Flagge..

Triage Decision Matrix

Signal Type	Magnitude/Context	Primary Owner	Escalation Level	Typical Resolution Path
Volume Delta (%)	> 2x tolerance (PROD, ..	DPO + EDE Domain	Sev-2	Confirm source complet..
Volume Delta (%)	Within tolerance but per..	EDE Domain	Sev-4	Baseline recalibration ta..
Null Ratio (%)	Exceeds tolerance on cr..	EDE Domain	Sev-3	Validate mapping; adjus..
Duplicate Rate (%)	Any exceedance on pri..	DPO + EDE Domain	Sev-3	De-duplication check; re..
Schema Change	Not linked to approved ..	EDE Domain	Sev-2	Rollback change; align ..
Key Violation	Repeated across 2 parti..	EDE Domain	Sev-2	Constraint verification; i..

Monthly Calibration Review (Volume/Schema)

Step	Activity	Stakeholders	Artifact Produced	Reference Tickets
1	Evaluate false positives/..	DPO, Observability Eng..	Calibration Summary (C..	PRB-1299, PRB-1321
2	Update baselines for str..	EDE Domain Leads	Baseline Parameter Ch..	CHG-98102, CHG-98322
3	Reassess Tier classifica..	Data Governance, DPO	Tiering Decision Memo ..	GOV-774
4	Validate catalog contrac..	EDE, Data Catalog Team	Contract Reconciliation ..	TSK-98001-98045
5	Publish KPI attainment ..	DPO	Monthly KPI Report (MK..	KPI-2026-07

Case Notes (Excerpts)

- INC-44498 (2026-07-11): Trades_Equities_T+0 volume dip detected at -12.8% vs weekday baseline. EDE-Equities confirmed late vendor segment S3; backfill job executed 17:05-17:22Z. m.freshness_minutes remained within SLA; volume normal by 17:25Z. Root cause tagged as "External Partial Delivery." PRB-1330 opened to observe 14-day trend.
- INC-44527 (2026-07-12): Custody_Balance_Snapshot duplicate rate measured at 0.36% on account_id. Suspected overlap from a manual retry proximate to success event. DPO annotated concurrency window; EDE-Custody reviewing idempotency checks. Interim suppression window extended to 60 minutes for this pipeline while investigation proceeds. No downstream aggregates affected (reconciliation totals unchanged).
- TSK-98256 (2026-07-12): Reference_Security_Master schema column "issuer_country_alpha3" added under CHG-98322. Schema monitor correctly registered change as informational; Data Catalog updated at 05:02Z. No consumer breakage reported.

Quality Indicators (Rolling 30 Days)

Indicator	Result	Target	Notes
Volume Anomaly Detection Pr..	93.8%	>= 92%	After baseline reweighting for ..
Schema Drift Detection SLA (t..	8 minutes median	<= 10 minutes	From detection to ticket annota..
Duplicate Rate Exceedances (..	3	<= 4	All attributed to retry overlap wi..
Null Ratio Exceedances (count)	6	<= 6	2 were accepted due to vendor..

Data Contracts and Catalog Alignment

- All schema drift alerts are cross-referenced to the Data Catalog (Catalog ID prefix DCAT-) and require an approved CHG ticket prior to runtime enforcement changes.
- Contract mismatches produce informational alerts for UAT, actionable alerts for PROD. DPO ensures annotation within 15 minutes of detection for Tier-1 feeds.

Stakeholder Communication

Audience	Trigger Condition	Communication Channel	Content Owner
BI Duty Manager	Tier-1 Sev-2 volume anomaly	Email DL: BI-Alerts, #bi-duty	DPO On-Call

Risk Analytics	Rates_Curve_Intraday null rati..	#risk-data	EDE-Rates
Data Governance	Unapproved schema change d..	Email: Data.Governance@	EDE Domain Lead

Retention and Auditability

- Anomaly detection results (scores, baselines, annotations) are preserved for 12 months at 5-minute resolution and summarized weekly for audit (AUD-DS-Volume-YYYYWW).
- All changes to thresholds or tiering require a CHG ticket and are reflected in the Baseline Parameter Change Log with effective dates.

The combination of seasonal baselines, explicit tolerances, and structured triage ensures that volume and schema drift signals are actionable while minimizing alert fatigue. DPO continues to partner with EDE and Observability Engineering to refine parameters where persistent benign deviations are observed without downstream impact.

Section 4. Alert Taxonomy, Thresholds, and Routing

Meridian Capital's ETL pipeline monitoring is governed by a shared taxonomy to ensure consistent detection, triage, and reporting across data domains. This section enumerates alert dimensions, severity thresholds, routing logic, and maintenance exceptions applied by Data Platform Operations (DPO) in partnership with Enterprise Data Engineering (EDE), Risk Data Services (RDS), and SRE Observability (SRE-O). All thresholds are calibrated to support business SLAs and data freshness SLOs without generating unnecessary noise. Changes to thresholds and routes require approval via the Data Reliability Change Advisory Board (DR-CAB).

Monitoring dimensions and associated metric identifiers are standardized to enable aligned dashboards and comparative analytics across pipeline families.

Table 4.1: ETL Monitoring Dimensions and Metric Identifiers

- Dimension: Schedule Adherence

Metric ID: etl.schedule.delay.minutes

Description: Minutes delayed vs. scheduled start

Collection Frequency: 1 min

Source System: Airflow Scheduler (af-scheduler-ct01)

- Dimension: Data Freshness

Metric ID: etl.data.freshness.lag_minutes

Description: Lag between last successful load and current time

Collection Frequency: 5 min

Source System: MDL Metadata Service (mdl-meta-02)

- Dimension: Task Reliability

Metric ID: etl.task.retry.count

Description: Count of retries per task instance within window

Collection Frequency: Per task completion

Source System: Airflow Worker Fleet (af-worker-pool-EAST)

- Dimension: Throughput

Metric ID: etl.records.processed.rate

Description: Records processed per minute

Collection Frequency: 1 min

Source System: Transport Layer Metrics (kafka-metrics-gw)

- Dimension: Error Rate

Metric ID: etl.errors.rate

Description: Errors per 1,000 records processed

Collection Frequency: 1 min

Source System: Log Ingestor (log-agg-01)

- Dimension: Load Duration

Metric ID: etl.stage.load.duration.sec

Description: End-to-end stage load duration

Collection Frequency: Per load

Source System: Job Orchestrator (orchestrator-svc)

- Dimension: SLA Breach

Metric ID: etl.sla.breach.flag

Description: Boolean flag when SLA is exceeded

Collection Frequency: Event-driven

Source System: SLO Evaluator (slo-eval-api)

Severity is assigned based on a combination of metric value, duration outside threshold, and business calendar context (trading vs. non-trading hours). The classification below maps to PagerDuty service routing and ServiceNow incident creation.

Table 4.2: Severity Matrix and Trigger Conditions

- Severity: SEV-1 (Critical)

Trigger Condition: etl.sla.breach.flag=1 for Tier-1 datasets OR etl.data.freshness.lag_minutes > 120 during trading hours

Observation Window: >=10 min continuous

Auto-Clear: Manual only

Example Alert ID: A-2024-0715-000481

- Severity: SEV-2 (High)

Trigger Condition: etl.task.retry.count >= 5 for any task in Tier-1 pipeline OR etl.errors.rate >= 10/1k >= 15 min

Observation Window: >=15 min or >=1 task window

Auto-Clear: Requires stable metrics for 30 min

Example Alert ID: A-2024-0715-000492

- Severity: SEV-3 (Moderate)

Trigger Condition: etl.schedule.delay.minutes >= 20 outside maintenance window OR throughput drops >= 40% vs. 7-day median for 30 min

Observation Window: >=30 min

Auto-Clear: Yes, upon recovery for 30 min

Example Alert ID: A-2024-0716-000031

- Severity: SEV-4 (Informational)

Trigger Condition: etl.records.processed.rate anomaly z-score >= 2 excluding known backfills

Observation Window: Point-in-time

Auto-Clear: N/A (one-shot)

Example Alert ID: A-2024-0716-000045

Routing for alerts is governed by severity, data domain, and time of day. Primary escalation flows are defined below. All SEV-1/SEV-2 alerts auto-create incidents in ServiceNow with linkage to the triggering metric and affected pipeline identifiers.

Table 4.3: Alert Routing Matrix

- Severity: SEV-1

Business Hours (08:00-18:00 ET): DPO Primary + EDE On-Duty

After Hours: DPO On-Call via PagerDuty Escalation Policy EP-ETL-01

Secondary Channel: #etl-major-incidents (Teams)

Ticket Auto-Create: Yes (INC)

Ticket Queue: SNOW Assignment Group: DPO-ETL

- Severity: SEV-2

Business Hours (08:00-18:00 ET): DPO Triage

After Hours: DPO On-Call

Secondary Channel: #etl-ops

Ticket Auto-Create: Yes (INC)

Ticket Queue: SNOW Assignment Group: DPO-ETL

- Severity: SEV-3

Business Hours (08:00-18:00 ET): DPO Triage

After Hours: Queued for next business day unless persistent >90 min

Secondary Channel: #etl-ops

Ticket Auto-Create: Yes (TASK)

Ticket Queue: SNOW Assignment Group: DPO-ETL

- Severity: SEV-4

Business Hours (08:00-18:00 ET): No page; dashboard only

After Hours: Suppressed

Secondary Channel: N/A

Ticket Auto-Create: No

Ticket Queue: N/A

Maintenance, backfills, and vendor blackout periods are codified as squelch windows to reduce false positives. Squelch windows require approval and are time-boxed with explicit ticket references.

Table 4.4: Approved Squelch Windows (as of 2026-07-01)

- Pipeline Group: CREDIT_RISK_DAILY

Window (ET): Sat 20:00-Sun 04:00

Scope: schedule delay, throughput anomalies

Rationale: Weekly compaction and index maintenance

Approval Ticket: CHG-ETL-68231

- Pipeline Group: CORE_BANKING_CDC

Window (ET): Daily 00:15-00:40

Scope: transient retry spikes

Rationale: Source DB log rollover

Approval Ticket: CHG-ETL-67712

- Pipeline Group: MARKET_FEEDS_INTRADAY

Window (ET): Trading holidays (NYSE calendar)

Scope: data freshness lag only

Rationale: No upstream feed

Approval Ticket: CHG-ETL-68104

The following audit log excerpt demonstrates a consolidated alert generated under the new taxonomy, with enrichment from the SLO Evaluator and Airflow metadata services. This example reflects a SEV-2 event for a Tier-1 pipeline during business hours.

Consolidated Alert Metadata (Excerpt)

- Timestamp (UTC): 2026-07-15T14:22:08Z
- Alert ID: A-2026-0715-000492
- Incident (ServiceNow): INC00192847
- Pipeline ID: PIPE-RDS-T1-CDC-CBK-L01
- Affected Task: transform_load_cbk_txn_hourly
- Severity: SEV-2
- Trigger Metric: etl.task.retry.count=6
- Observation Window: 2026-07-15T14:05:00Z-2026-07-15T14:20:00Z
- Business Calendar: TradingDay
- Routing: EP-ETL-01 (DPO On-Call), Slack #etl-ops bridge
- Related Changes: CHG-ETL-68177 (airflow provider upgrade)
- Owner Group: DPO-ETL
- Acknowledged By: dpo_oncall (14:22:35Z)
- Auto-Clear Policy: stabilize 30 min; manual closure required if INC created

Change history for thresholds and routes is tracked to provide context during incident review. Recent threshold updates are summarized below.

Table 4.5: Threshold and Routing Change Log

- Change Ref: CHG-ETL-68177

Date/Time (ET): 2026-07-12 22:30

Scope: Increased etl.task.retry.count SEV-2 trigger from >=4 to >=5 for Tier-1

Requestor: EDE-Core (jchen)

Approver: DR-CAB (meeting 2026-07-12)

Post-Change Review Ticket: PRB-ETL-11934

- Change Ref: CHG-ETL-68198

Date/Time (ET): 2026-07-13 19:00

Scope: Route SEV-3 schedule delays to TASK queue only

Requestor: DPO (smalik)

Approver: DR-CAB (asynchronous)

Post-Change Review Ticket: PRB-ETL-11939

Dashboard alignment: Grafana folders "ETL/Service Health" and "ETL/SLOs" reflect the above identifiers and severity mapping. Panels are labeled with Metric IDs to ensure that screen captures in incident records are unambiguous. SRE-O RVT-2026-Q3 verified panel queries align with metric naming; see verification ticket RVT-OBS-423 for evidence and peer review.

Operational Note: All severity evaluations must consider approved squelch windows. If a SEV-2/SEV-1 triggers during a squelch, the alert is suppressed and a log annotation is created with reference to the change ticket and reason code. Suppression logic is auditable via the Alert Suppression Report (run weekly; ticket RPT-ETL-SQUELCH-W28).

Section 5. Runbook Procedures for Common Alerts

This section provides standardized, non-destructive operational procedures for the most frequent alert categories. Each runbook includes the expected signal, key parameters, target handling times, and post-resolution activities. These procedures are designed for DPO triage with collaboration from EDE and data domain owners as indicated. All times are in Eastern Time unless otherwise specified.

Runbook 5.1: Data Freshness Lag Breach (Tier-1 Daily Aggregate)

Context: Triggered when etl.data.freshness.lag_minutes exceeds 120 minutes for Tier-1 datasets during trading hours. Typical drivers include upstream vendor delays, protracted stage loads, or prolonged backfill windows without active squelch.

Alert Parameters (Example: A-2026-0716-000031)

- Dataset: CREDIT_RISK_DAILY (crd_risk_daily_vw)
- Pipeline ID: PIPE-RDS-T1-BATCH-CRD-D01
- Severity: SEV-1 during 08:00-18:00 ET
- First Seen: 2026-07-16T09:35:02Z
- Latest Evaluated Lag: 137 minutes
- Last Successful Load: 2026-07-16T07:18:00Z
- Related Change Window: None active
- Business Owner: Risk Analytics (ra_ops)

Resolution Steps (Target: Acknowledge <=5 min; Mitigate <=45 min)

- Step: 1

Actor: DPO

System/Tool: PagerDuty, ServiceNow

Action: Acknowledge alert; confirm INC created and assign to DPO-ETL

Expected Outcome: Incident INC00193104 in "In Progress"

Time Budget: 5 min

- Step: 2

Actor: DPO

System/Tool: Grafana "ETL/SLOs"

Action: Validate lag metric trend and confirm Tier-1 classification

Expected Outcome: Lag >120 min sustained >=10 min; Tier-1 tag present

Time Budget: 5 min

- Step: 3

Actor: DPO

System/Tool: Airflow UI (read-only), Pipeline Metadata

Action: Review last successful run status and current running tasks

Expected Outcome: Identify stage/task currently on critical path

Time Budget: 10 min

- Step: 4

Actor: DPO with EDE

System/Tool: Job Orchestrator, MDL Metadata

Action: Check for active backfill or squelch; if none, coordinate with EDE to assess task queue health and dependencies

Expected Outcome: Determination if delay is upstream vs. internal

Time Budget: 10 min

- Step: 5

Actor: DPO

System/Tool: Stakeholder Channel (#risk-data-ops)

Action: Notify business owner of expected recovery ETA based on findings

Expected Outcome: Stakeholder acknowledgment recorded in INC work notes

Time Budget: 5 min

- Step: 6

Actor: EDE

System/Tool: Orchestrator controls (within change boundaries)

Action: If safe, reprioritize current run within queue to recover SLA; otherwise provide recovery ETA based on resource availability

Expected Outcome: Updated ETA noted; lag begins to contract

Time Budget: 10 min

Post-Resolution Activities

- Task: Confirm freshness restored (lag <30 min sustained 30 min)

Owner: DPO

Evidence: Grafana screenshot attached to INC

Due: Within 1 hr of recovery

- Task: Root cause categorization (Upstream vs. Internal vs. Mixed)

Owner: EDE

Evidence: PRB ticket if repeated within 7 days

Due: 1 business day

- Task: SLO impact annotation

Owner: DPO

Evidence: Update in SLO Evaluator notes (slo-eval run ID)

Due: At incident closure

Runbook 5.2: Elevated Retry Count on Transform Task

Context: Triggered when etl.task.retry.count >= 5 for any task in Tier-1 pipelines. Common contributors include transient upstream schema drifts flagged by validations, partition skew, or intermittent transport layer disruptions. This runbook focuses on confirming stability and coordinating remediation within approved operational controls.

Alert Parameters (Example: A-2026-0715-000492)

- Pipeline ID: PIPE-RDS-T1-CDC-CBK-L01
- Task: transform_load_cbk_txn_hourly
- Severity: SEV-2
- Retries Observed: 6 in 15 min window
- Last Successful Task Start: 2026-07-15T13:58:42Z
- Maintenance Window: Not active
- Related Changes: CHG-ETL-68177 (provider upgrade completed 2026-07-12)

Resolution Steps (Target: Acknowledge <=5 min; Stabilize <=30 min)

- Step: 1

Actor: DPO

System/Tool: PagerDuty, ServiceNow

Action: Acknowledge alert; verify incident INC00192847 linkage

Expected Outcome: Assignment confirmed to DPO-ETL

Time Budget: 5 min

- Step: 2

Actor: DPO

System/Tool: Airflow UI (read-only)

Action: Open task instance details; check retry interval adherence and logs summary for non-sensitive validation messages

Expected Outcome: Retry pattern matches orchestration policy; no runaway loop

Time Budget: 5 min

- Step: 3

Actor: DPO with EDE

System/Tool: Grafana "ETL/Service Health"

Action: Correlate retries with throughput and error rate panels

Expected Outcome: Either localized task issue or broader platform signal

Time Budget: 10 min

- Step: 4

Actor: EDE

System/Tool: Orchestrator controls

Action: If retries are stabilizing and last attempt succeeded, allow natural completion; if still escalating, coordinate a controlled task rerun within change boundaries

Expected Outcome: Retries cease; task reaches success or deterministic failure path

Time Budget: 10 min

Escalation and Communication

- If SEV-2 persists >30 min without improvement, page EDE On-Duty (PD schedule: EDE-Core) and notify #etl-ops with current hypothesis and ETA.
- If multiple pipelines show similar patterns, notify SRE-O to assess shared components (ref: bridge call code BRG-ETL-2026-07-15-A).

Closure Criteria and Documentation

- Condition: Retry count returns to baseline (<=1) for subsequent runs within 2 hours.
- Documentation: Attach panel exports and Airflow task summary to INC; set resolution code "Stabilized/No Change Required" or "Mitigated/Operational Adjustment."

Runbook 5.3: Schedule Delay Outside Approved Window (SEV-3)

Context: Triggered when etl.schedule.delay.minutes >= 20 outside defined squelch windows. Typical during high-load batch hours when queue contention occurs.

Alert Parameters (Example: A-2026-0716-000045)

- Pipeline Group: FINANCE_MONTHLY
- Affected DAG: agg_fin_monthly_close
- Scheduled Start: 2026-07-16T21:00:00Z
- Actual Start: 2026-07-16T21:23:00Z
- Severity: SEV-3
- Squelch: Not applicable (window Sat/Sun only)

Resolution Steps (Target: Acknowledge <=10 min; Plan <=30 min)

- Step: 1

Actor: DPO

System/Tool: ServiceNow, Grafana

Action: Confirm delay; check concurrent DAG count vs. pool capacity

Expected Outcome: Identified queue contention or transient delay

Time Budget: 10 min

- Step: 2

Actor: DPO with EDE

System/Tool: Orchestrator pools

Action: If safe, adjust run priority for Tier-1 workloads; document in INC

Expected Outcome: Start time normalized within next cycle

Time Budget: 20 min

Post-Resolution Activities (All Runbooks)

- Metric Capture: Record "time to acknowledge," "time to mitigate," and "time to recover" in the Incident Metrics tab.
- Knowledge Update: If procedure deviated materially, create KB draft and link to incident (KB-ETL-Draft-2026-07-XX).
- Review Cadence: Include in weekly Ops Review (WOR-ETL-Week28) with action items tracked in TASK-ETL-Backlog.

Appendix 5.A: Example Incident Work Notes (Condensed)

- 2026-07-15 10:22 ET, DPO (smalik): Acknowledged A-2026-0715-000492; INC00192847 opened; paging EDE On-Duty.
- 2026-07-15 10:34 ET, EDE (jchen): Observed retries stabilizing; last attempt succeeded; monitoring next run.

- 2026-07-15 10:58 ET, DPO (smalik): No further retries; closing PD; keeping INC open for 30-min observation.
- 2026-07-15 11:30 ET, DPO (smalik): Stable. Attached Grafana panel exports. Resolving INC with code "Stabilized."

Quality Controls and Review

- Quarterly Validation: SRE-O conducts alert drift assessment (RVT-OBS-423) to recalibrate thresholds against rolling 90-day baselines.
- Business Impact Check: RDS confirms that Tier-1 data sets met SLO coverage >=99.5% per month; exceptions logged in SLO-EXC-2026Q3.
- Ticket Hygiene: All incidents must include a link to at least one screenshot or metric panel and a stated hypothesis/outcome.

End of Sections 5.1-5.3. Subsequent sections detail dependency health indicators and cross-system correlation panels used during complex, multi-pipeline triage events.

Section 4. Alert Optimization and Review (continued)

4.1 Quarterly Alert Noise Review (Q3 FY26) -- Meridian Capital Data Platforms

Summary

On 2026-07-19 at 10:30-11:15 ET, the Platform Observability Guild (POG), Enterprise Data Engineering (EDE), and Business Operations Analytics (BOA) convened for the scheduled Q3 alert noise review (Ref: MTG-ETL-ALERT-2026Q3). The primary objective was to evaluate alert volume, quality, and operational impact for core ETL pipelines, with emphasis on ingestion latency and load failure alerts in PROD. All analyses reflected the period 2026-06-17 to 2026-07-16 inclusive.

Participants

- POG: A. Khanna (Chair), K. Montero
- EDE: J. Lin (ETL RunOps Lead), M. Sato (Data Reliability)
- BOA: D. Connors (Analytics Ops)
- Change Mgmt: Observer -- S. Hsu

Alert Definition Reference (current state)

- Name: ETL Ingestion Latency High
- ID: ALERT-ETL-INGEST-LATENCY-HIGH
- Scope: PROD, pipelines tagged etl_core=true
- Metric: p95 end-to-end ingest latency (source receipt to curated zone write)
- Trigger Threshold: p95 > 12 minutes for 15-minute rolling window
- Evaluation Cadence: 1-minute intervals
- Data Volume Gate: not set (current); any volume
- Maintenance Window Suppression: 02:15-02:45 ET (Sunday) planned release window
- Owner Group: EDE RunOps
- Escalation Policy: EP-ETL-02 (On-Call -> Secondary -> Duty Manager)
- Runbook Link: KB-ETL-0012
- Created: 2026-03-04 (CHG-2026-0317)
- Last Updated: 2026-06-12 (CHG-2026-1083) to add tag etl_core=true

Findings (Q3-to-date)

- Alert Volume: 412 total events (PROD); 38 paged; 51 acknowledged within 5 minutes without paging.
- Mean Time to Acknowledge (MTTA): 3m 42s; Median: 1m 58s.
- False Positive Estimate: 27% (based on post-event classification where SLA not at risk).
- Top 3 Noisy Alerts by Dataset Context:

1) ledger_txn_daily -- 92 events; 21% associated with planned schema maintenance windows.

2) client_ref_delta -- 61 events; 35% due to low-volume off-peak ingestion skew.

3) risk_model_factors -- 54 events; 18% correlated with upstream late file arrivals (non-actionable locally).

- Notable Period: 2026-06-17 (release train RT-ETL-26.24). Temporary spike (31 events in 2 hours) coincident with metadata crawler backoff; noise attributed to unadjusted suppression window.

Metrics Snapshot (Week of 2026-07-12)

- Period: 2026-07-12 00:00 ET to 2026-07-18 23:59 ET
- Total Alerts (ALERT-ETL-INGEST-LATENCY-HIGH): 69
- Paged Events: 7
- Auto-Closed (self-resolved within 5 min): 19
- Average Duration to Return Below Threshold: 6m 14s
- Incidents Opened (INC): 3 (INC-2026-07911, INC-2026-07928, INC-2026-07957)
- Problem Records Initiated (PRB): 1 (PRB-2026-0041 -- client_ref_delta skew)

Proposed Tuning (to reduce non-actionable noise)

- Change Request: CHG-2026-1189 (scheduled 2026-07-25 CAB)
- Revisions:
  - Increase trigger threshold to p95 > 15 minutes (maintains SLA coverage; reduces off-peak noise).
  - Introduce data volume gate: minimum 5,000 records per 15-minute window to fire alert.
  - Expand planned maintenance suppression: Sundays 02:00-03:00 ET.
  - Add correlation tag: upstream_source_status to assist in auto-classification.
  - Limit scope to etl_core=true AND env=PROD; remove UAT spillover.
- Risk Assessment: Low. Historical backtest suggests 23% event reduction with no missed SLA breaches.
- Rollback Plan: Revert to ALERT-ETL-INGEST-LATENCY-HIGH@v3 within 10 minutes on adverse signal (Ref: KB-ETL-0012 Sec.5.2).
- Owner: A. Khanna (POG)

Operational Log Excerpt (annotated)

- 2026-07-15 08:02:14 ET -- Event E-882217 fired (ledger_txn_daily). Paged EDE On-Call (EP-ETL-02). Ack 08:04:01 (Ref: ACK-ETL-3319).
- 2026-07-15 08:06:33 ET -- Correlated upstream tag upstream_source_status=delayed_files. Opened INC-2026-07928 for tracking. No user impact note posted (no SLA risk).
- 2026-07-15 08:13:02 ET -- p95 returned to baseline (8m 11s). Event auto-resolved at 08:13:15. Root-cause: scheduled schema evolution for partition pruning; added analyze overhead (~8m).
- 2026-07-15 08:23:40 ET -- Action recorded: add maintenance tag for schema evolution tasks (KB-ETL-0012 update submitted; DOC-2026-0417). No recurrence observed next window.

Noise Drivers -- Root Themes and Remediations

- Off-peak volume skew resulted in volatility in p95 for low-volume datasets. Proposed data volume gate directly addresses this.
- Misaligned suppression windows during release trains created avoidable paging. CAB-aligned suppression extension proposed.
- Upstream late arrivals frequently triggered local alerts without local remedies. Tagging and auto-classification reduce manual triage cycles.

Daily Alert Hygiene Checklist (EDE RunOps)

Step 1

- Action: Review top 5 alerting datasets over last 24h; confirm linkages to incidents or maintenance.
- Owner: Shift Lead (EDE)

- Expected Outcome: All repeated alerts have associated ticket references.

Step 2

- Action: Validate active mutes/suppressions vs. release calendar (CAL-ETL-REL-2026).
- Owner: POG
- Expected Outcome: No mutes outside approved windows.

Step 3

- Action: Confirm runbook links and context tags present on all firing alerts.
- Owner: EDE RunOps
- Expected Outcome: >=98% alerts with KB references and dataset tags.

Step 4

- Action: Sample 3 raw events for correlation accuracy (upstream_source_status, dataset).
- Owner: POG
- Expected Outcome: Tag accuracy >=95%.

Step 5

- Action: Record hygiene status in Daily Ops Note.
- Owner: Shift Lead
- Expected Outcome: Note created with timestamp and exceptions logged.

Action Items and Ownership

- CHG-2026-1189 preparation -- POG (A. Khanna) by 2026-07-22.
- KB-ETL-0012 section on volume gating -- EDE (M. Sato) by 2026-07-24.
- BOA review of business impact mapping for client_ref_delta -- BOA (D. Connors) by 2026-07-29.

Section 4.2 ETL SLA Breach Triage Playbook (Non-Outage)

Scope and Intent

This playbook defines the standard triage sequence when an ETL pipeline exhibits elevated latency without hard failure, potentially risking but not yet breaching daily service-level commitments. It applies to PROD ETL workloads tagged etl_core=true and data products with published SLAs (Ref: SLO-ETL-2026). The playbook is designed to minimize time-to-diagnosis while ensuring consistent stakeholder communication.

Entry Criteria

- Alert: ALERT-ETL-INGEST-LATENCY-HIGH fired in PROD AND
- SLA Window: within 120 minutes of an SLA checkpoint OR
- Observed: p95 latency > threshold for >=15 minutes

Exit Criteria

- p95 latency stabilized below threshold for two consecutive windows OR
- Controlled mitigation in place and downstream stakeholders notified OR
- Incident escalated to Major Incident process (not covered here)

Standard Triage Steps (RunOps)

Step 1

- Action: Acknowledge alert; confirm dataset(s), job ID(s), and environment.
- Owner: EDE On-Call
- Target Time: <=5 minutes
- Output: Incident opened if not existing (INC-YYYY-XXXXX).

Step 2

- Action: Review ETL Latency Overview (PROD) -- Dashboard ID DB-ETL-OVW-03.
- Owner: EDE On-Call
- Target Time: <=7 minutes
- Output: Identify whether impact is localized (single dataset/job) or systemic.

Step 3

- Action: Correlate upstream indicators (upstream_source_status, arrival_count_last_15m).
- Owner: EDE On-Call with POG support
- Target Time: <=10 minutes
- Output: Classification tag applied: upstream_delay | capacity_pressure | metadata_ops.

Step 4

- Action: Check work queue depth and recent batch counts for affected pipeline(s).
- Owner: EDE On-Call
- Target Time: <=10 minutes
- Output: Determine if queuing is increasing, stable, or clearing.

Step 5

- Action: Validate maintenance/release windows in CAL-ETL-REL-2026.
- Owner: POG
- Target Time: <=5 minutes
- Output: Maintenance correlation confirmed or ruled out.

Step 6

- Action: Post initial status note to stakeholders.
- Owner: Incident Owner (EDE)
- Target Time: <=20 minutes from alert
- Output: Status includes scope, preliminary cause, next update timebox.

Step 7

- Action: Implement low-risk mitigations aligned with runbook KB-ETL-0012 (e.g., queue prioritization for at-risk dataset).
- Owner: EDE On-Call
- Target Time: <=25 minutes
- Output: Mitigation status recorded in incident.

Step 8

- Action: Monitor p95 trend for two consecutive windows.
- Owner: EDE On-Call
- Target Time: Ongoing
- Output: Stabilization confirmed or further action required.

Step 9

- Action: If latency persists and SLA checkpoint approaches (<60 minutes), engage Secondary and Duty Manager per EP-ETL-02.
- Owner: EDE On-Call
- Target Time: As threshold approaches
- Output: Escalation noted in incident.

Step 10

- Action: Upon stabilization, record timeline, cause classification, and any adjustments applied.
- Owner: Incident Owner
- Target Time: Within 60 minutes of resolution
- Output: Incident closure summary.

Escalation Matrix Reference

- Policy: EP-ETL-02
- Primary On-Call: EDE RunOps (24x7)
- Secondary: Data Reliability Engineer (06:00-22:00 ET)
- Duty Manager: Data Platforms Duty (on rotation)

- Advisory (as needed): Platform Observability Guild; Business Data Steward for impacted product

SLA Context Examples

- ledger_txn_daily: SLA 08:00 ET data availability; warning threshold at 06:30 ET if p95 > 15m.
- client_ref_delta: SLA 07:30 ET; escalate if p95 > 15m with arrival_count_last_15m < expected baseline.
- risk_model_factors: SLA 09:00 ET; advisory to Risk Analytics if at-risk window detected before 08:30 ET.

Case Study (Non-Outage)

- Ticket: INC-2026-08192
- Date: 2026-08-02
- Dataset: ledger_txn_daily
- Trigger: ALERT-ETL-INGEST-LATENCY-HIGH at 06:51 ET; p95 at 17m over 15-minute window
- Timeline:
  - 06:52 ET -- Ack by EDE On-Call; incident created.
  - 06:58 ET -- Dashboard DB-ETL-OVW-03 indicates localized queue buildup; upstream_source_status=delayed_files.
  - 07:06 ET -- Queue prioritization applied to ledger_txn_daily batch ID B-2026-08-02-01 (per KB-ETL-0012 guardrails).
  - 07:21 ET -- p95 trending down; queue clearing confirmed.
  - 07:43 ET -- p95 stabilized at 9m; SLA no longer at risk.
- Outcome: No breach. Root cause attributed to upstream late file delivery. PRB-2026-0046 opened to improve coordination and early-warning signalization.
- Follow-up: BOA alignment on stakeholder notification thresholds. Preventive item: PRJ-ETL-112 to refine baseline arrival models.

Key Performance Indicators (Triage Efficiency)

- MTTA (Mean Time to Acknowledge): target <=5m (last 30 days: 3m 18s).
- MTTDIA (Mean Time to Diagnosis): target <=15m (last 30 days: 12m 04s).
- MTTMIT (Mean Time to Mitigation): target <=25m (last 30 days: 21m 36s).
- Communication SLA: initial status <=20m; cadence every 30m while at risk (last 30 days: 98% adherence).

Triage Data Capture Fields (for Incident Forms)

- Incident ID
- Dataset/Product
- Job ID(s)
- Environment
- Observed Metric and Value
- Threshold at Time of Fire
- Start Time (first symptom)
- Acknowledge and Resolution Timestamps
- Cause Classification Tag
- Upstream Correlation (arrival_count_last_15m, source status)

- Queue Depth Snapshot (count, trend)
- Actions Taken (summary)
- Stakeholders Notified

Artifacts to Attach

- Two screenshots from DB-ETL-OVW-03 (before/after trend)
- Excerpt of event annotations (correlation tags, dataset)
- Notes from maintenance calendar review (if applicable)

Change Management Linkages

- CHG-2026-1205 (post-review) -- approved minor adjustment to batch prioritization policy for at-risk datasets. Performance validated by POG; no regression observed in the week following change.
- KB Updates: KB-ETL-0012 Sec.6.1 expanded to clarify classification tags and status note structure.

Communication Template (Internal Status Note)

- Primary Audience: EDE distribution list, Data Steward of impacted product, BOA liaison.
- Minimum fields: Incident ID, scope, current p95, cause classification, next update time, SLA checkpoint proximity.
- Example: "INC-2026-08192 | ledger_txn_daily | p95 17m (threshold 15m) | cause: upstream_delay | mitigation: prioritized queue | next update 07:20 ET | SLA 08:00 ET."

This playbook is reviewed semi-annually. Next review: 2027-01-15 (CAB schedule permitting). Owner: EDE RunOps Lead (J. Lin).

ETL Pipeline Monitoring and Alerting Handbook

Meridian Capital

Pages 11-12

Page 11: Alert Routing, Escalation, and Noise Reduction

Overview

This page documents how ETL monitoring alerts are routed, acknowledged, and escalated within Meridian Capital, and summarizes Q3 alert-noise reduction outcomes. Content reflects established operational practices administered by Data Platform Operations (DPO) in close coordination with Enterprise Data Engineering (EDE), Finance Data Stewardship (FDS), and the Service Management Office (SMO). References are provided for auditing and continuity.

Alert Routing Matrix (Effective: 2026-07-01 00:00 UTC; Ref: DOC-ETL-MON-AR-11.2)

- Context: All alerts originate from the enterprise observability platform and are normalized to the ETL domain taxonomy before delivery to queues. Deduplication and severity normalization occur at ingest.

Alert Routing Parameters (Extract)

Source System	Condition Summ..	Severity	Primary Queue	Escalation (Time)	Escalation To	Channels	Suppression Win..
Airflow DAGs	DAG failure on fina..	P2	DPO-OnCall	15 min unacked	EDE-Primary	Pager, Email DL: e..	02:00-03:00 Sat C..
Data Quality	Critical test fail (dbt..	P2	FDS-Queue	20 min unacked	DPO-OnCall	Pager, Teams Cha..	None INC-452199
Kafka Ingestion	Consumer lag > ba..	P3	DPO-OnCall	60 min no improve..	Integration Service..	Email	00:00-01:00 Wed ..
File Watcher	Daily file missing T..	P2	FDS-Queue	30 min unacked	Source App Owner..	Email	Holiday Calendar ..
Warehouse Load	Rowcount delta < -..	P3	EDE-Primary	2 hrs no RCA note	FDS-Queue	Teams	None PRB-017518
Synthetics	Probe mismatch o..	P3	DPO-OnCall	90 min persistent	EDE-Secondary	Email	23:00-23:30 daily ..

Routing Notes

- Severity normalization harmonizes vendor severities into P1-P4 where P1 is reserved for cross-domain impact. ETL currently operates P2-P4 only.
- Suppression windows align to planned maintenance defined in the SMO change calendar; see CHG references for windows that impact DAG-level alerts.

Alert Lifecycle States and Targets (Ref: SOP-ETL-ALC-04)

- New: Generated, awaiting triage.
- Acknowledged: Owner identified; timer for escalation paused.
- In Progress: Active diagnostic or remediation underway; status note entered.
- Resolved: Alert condition cleared; post-incident note linked if required.
- Closed: SMO review complete; metrics captured.

Service Level Targets (Q3 Baseline)

Severity	Target MTTA (min)	Target MTTR (hrs)	Owner on First Ackno..	Reporting Tag
P2	10	4	DPO-OnCall	etl_core_prod
P3	30	8	DPO-OnCall or EDE-Pri..	etl_core_noncrit
P4	120	24	Team of Origin (as assi..	etl_informational

Q3 Noise Reduction Review (2026-07-15 14:30 UTC; Facilitator: DPO; Minutes: TASK-889104)

- Objective: Reduce actionable-alert volume without loss of signal; maintain MTTA for P2.
- Outcome Summary (to date):
  - Alert volume reduced 22% (June to July) with stable P2 MTTA at 8.4 min (target: 10).
  - False-positive rate for P3 decreased from 31% to 17% following baseline calibration for weekday/holiday seasonality and deduplication improvements.
  - No missed incidents attributable to suppression or threshold changes per SMO audit PRB-017640.

Tuning Changes Implemented (Extract; all changes approved via SMO)

Change Ref	Applied (UTC)	Control Type	Scope	Owner	Observed Effect (2 w..
CHG-0062843	2026-07-02 02:10	Threshold adjust	Airflow DAG retry cou..	DPO	13% fewer flaps; no in..
CHG-0062991	2026-07-07 23:45	Dedup window	Kafka lag anomaly P3	DPO	Alert bursts collapsed;..
CHG-0063057	2026-07-10 01:30	Seasonality model	Rowcount weekday ba..	EDE	False positives reduce..
CHG-0063110	2026-07-12 04:00	Suppression rule	Synthetics during kno..	DPO/FDS	Eliminated known plan..

Incident Performance Snapshot (2026-07-01 through 2026-07-15; Ref: RPT-ETL-KPI-0715)

Severity	Count	MTTA (min)	MTTR (hrs)	SLA Met (Yes/No)	Notes
P2	12	8.4	3.6	Yes	Two events auto-clear..
P3	41	21.7	6.9	Yes	7 flagged as informati..
P4	19	67.3	12.4	Yes	No customer-facing im..

Case Note: ETL-TXN-AGG-04 Alert Consolidation (Ref: PRB-017518; Window: 2026-07-08 00:00-06:00 UTC)

- Summary: Nightly rowcount-delta P3 alerts triggered on partitioned aggregates for trading transactions. Seasonality adjustment and deduplication reduced 4 alerts to 1 per cycle without delaying real anomalies.
- Observations:
  - Partition variance elevated on settlement EOM; baseline update applied using D-28 rolling median with weekday pairing.
  - No material change to MTTR; analyst triage time per alert reduced from 12 min to 4 min.

Event Log Excerpt (Correlated)

Timestamp (UTC)	Actor	System	Action Summary	Ticket/Note
2026-07-08 01:14:22	AlertEngine	Observability	Generated P3 rowcount..	ALR-944201
2026-07-08 01:16:03	DPO-OnCall	ITSM	Acknowledged; tagged "..	INC-452611
2026-07-08 01:39:40	EDE-Analyst	Dashboard	Baseline graph reviewe..	TASK-887991
2026-07-08 02:05:17	DPO-OnCall	ITSM	Linked to PRB; request..	PRB-017518
2026-07-10 01:31:10	DPO/EDE	Observability	Applied CHG-0063057; ..	CHG-0063057
2026-07-12 06:03:49	SMO	ITSM	Closed PRB; audit trail ..	PRB-017518

Ownership and Contact Windows

- DPO primary on-call: 24/7, follow-the-sun rotation; handoff 07:00 UTC and 19:00 UTC.
- EDE primary support: 06:00-22:00 America/New_York on business days; outside hours via EDE-Secondary.
- FDS coverage: 08:00-20:00 America/New_York on business days; exceptions per SMO change calendar.

Documentation and References

- Handbook governance: RVP-Data Platform (Sponsor), Control ID: CTRL-ETL-MON-0011.
- Record of Change for this page: CHG-0063199 (2026-07-15 15:02 UTC) editorial update to routing matrix terminology.

Page 12: Metric Baselines, Data Freshness, and Synthetics

Overview

This page details how ETL monitoring baselines are established and evaluated, how data freshness is measured against service objectives, and how synthetic checks validate critical data paths in Meridian Capital's data ecosystem. The approach emphasizes stable detection, seasonality awareness, and business-aligned thresholds.

Baseline Configuration Principles (Ref: STD-ETL-MET-02)

- Rolling Window: 28-day lookback aligned by weekday to accommodate trading and finance calendars.
- Aggregation: Median and interquartile range (IQR) used to reduce sensitivity to spikes.
- Deviations: Anomalies flagged when observation deviates beyond dynamic bands (IQR*1.5 or 3sigma proxy) sustained over a defined duration.
- Seasonality Adjusters: U.S. market holidays and month-end/quarter-end markers applied to prevent predictable spikes from generating noise.

Baseline Parameters (Selected Metrics)

Metric ID	Metric Name	SLI Definition	Baseline Window	Deviation Rule	Min Observations	Owner	Ref
MET-ROW-001	Warehouse Rowco..		Deltarows(today vs..	/ max(D-7,1)	28D weekday-align..	Flag if < -40% for >..	10 EDE SOP-ETL-..
MET-LAT-004	Pipeline Latency	End-to-end duratio..	28D weekday-align..	Flag if > median + ..	15	DPO	RPT-ETL-KPI-0715
MET-LAG-010	Kafka Consumer L..	Max lag across par..	28D rolling	Flag if > baseline +..	20	DPO	PRB-017391
MET-DQ-020	Critical DQ Failure ..	(# critical fails / tot..	28D rolling	Flag if > 2% sustai..	20	FDS	DQ-ANX-2026-Q3
MET-FILE-030	File Arrival Jitter		actual - expected	arrival deviation (m..	28D weekday-align..	Flag if > 90 min (P..	12 FDS MC-ETL-2..

Data Freshness Objectives (Ref: SLO-DATA-FRESH-01; Review: 2026-07-10 16:00 UTC)

- Scope: Critical subject areas supporting Finance Close, Risk Analytics, and Regulatory Reporting.
- Measurement: Data watermark timestamps (ingestion_ts, as-of dates) compared to wall-clock, normalized by expected delivery cadence.
- Reporting: 95th percentile freshness assessed daily; exceptions logged to ITSM with dataset tags.

Freshness Targets and Thresholds

Dataset Key	Expected Arrival (Lo..	SLO (P95 Freshness,..	Alert Thresholds	Primary Owner	Notes
dw_core.accounts	01:30 America/New_Y..	<= 60	P2 if > 120; P3 if > 75	EDE	Driven by nightly acco..
dw_fin.ledger_postings	03:00 America/New_Y..	<= 90	P2 if > 180; P3 if > 120	FDS	Month-end variance to..
dm_risk.pnl_daily	07:00 America/New_Y..	<= 45	P2 if > 90; P3 if > 60	Risk Eng	Trading day alignment
stg_payments.settlem..	00:30 America/New_Y..	<= 30	P2 if > 60; P3 if > 45	EDE	Sensitive to bank holid..
dm_sales.pipeline_sn..	06:00 America/New_Y..	<= 120	P2 if > 240; P3 if > 150	EDE	Non-critical; business ..

Synthetic Path Monitoring (Ref: SYN-ETL-CHECKS-03)

- Purpose: Validate that critical data paths remain functional regardless of upstream variability by probing minimal, representative flows (e.g., schema-access checks, lightweight count probes, checksum validations on reference test files).
- Success Criteria: Probes confirm end-to-end observability (path reachable, transformations responsive, minimal artifact presence as expected) within the allowed window.
- Governance: Synthetic checks are non-intrusive and operate against designated test artifacts and views

maintained for monitoring; no production data is modified by these probes.

Synthetic Checks Schedule (Extract)

Check ID	Target Path/Data..	Probe Type	Frequency	Success Criteria (..	Ticket Trigger	Owner
SYN-AC-01	stg/accounts_inbo..	File checksum	15 min	Test file checksum..	P3 on 2/3 failures	DPO
SYN-LD-05	dw_core.accounts	Row existence	30 min	Min rowcount > 0 ..	P3 if 2 consecutive	EDE
SYN-DQ-02	dm_risk.pnl_daily	Null-rate probe	60 min	Null rate in key col..	P3 if sustained 60 ..	Risk
SYN-META-07	catalog registry	Metadata lookup	60 min	Dataset metadata ..	P4 on latency > 10s	DPO
SYN-LG-03	kafka/topic: trades..	Lag visibility	10 min	Lag metric endpoi..	P3 if missing 2 polls	DPO

Correlation and Anomaly Windows (Applied: 2026-07-12 00:00 UTC; CHG-0063110)

- Correlation Groups: Accounts ingestion (file arrival, rowcount delta, DQ critical rate) and Risk P&L (source extract, DAG latency, null-rate probe) now correlate within 60-min windows to consolidate multi-symptom events into a single incident stream.
- Expected Effect: Reduction in event fan-out where a single upstream delay would previously trigger multiple independent alerts.

Weekly Freshness Audit (Audit Window: 2026-07-05-2026-07-11; Report: RPT-FRESH-0712)

- Findings:
  - dw_fin.ledger_postings exceeded P3 threshold on 2026-07-08 (delay 137 min; within P2 threshold). Root cause: upstream source rescheduled per SMO CHG-0063022; exception noted; no remediation required.
  - dm_risk.pnl_daily maintained SLO at P95=37 min; one transient synthetic probe miss correlated with telemetry lag, not data unavailability.
  - stg_payments.settlements met P95=24 min; holiday calendar offset applied successfully on 2026-07-05.

Metrics Governance and Dashboards

- Master Dashboards: ETL Core Overview (Dash-ID: DP-ETL-Core), Data Freshness (Dash-ID: DP-Freshness-01), Synthetic Health (Dash-ID: DP-Synth-02).
- Ownership: DPO curates platform-level visuals; EDE and FDS maintain dataset- and pipeline-specific panels.
- Review Cadence: Weekly review on Tuesdays 15:00 UTC; action items logged to TASK queue with tags etl_metric_tuning and etl_slo.

Change Calendar Coordination (Extract; Source: SMO Calendar 2026-W29)

CHG ID	Window (UTC)	Impacted Pipelines	Risk Level	Monitoring Adjustme..	Comms Owner
CHG-0063190	2026-07-16 02:00-03:..	acct_load_daily_v3 (E..	Low	Suppress synthetics 0..	DPO
CHG-0063215	2026-07-17 23:00-23:..	ledger_postings_v2	Medium	Elevate freshness thre..	FDS
CHG-0063244	2026-07-18 04:00-05:..	risk_pnl_daily_v4	Medium	Extend correlation win..	Risk Eng
CHG-0063267	2026-07-19 01:30-02:..	settlements_ingest_v2	Low	Pause lag probe durin..	DPO

Quality Controls and Auditability

- All monitoring parameter adjustments require SMO-approved change records with backout criteria and post-change observation notes in ITSM.
- Baseline recalculations are logged with metric IDs, timestamp, initiator, and before/after summary statistics in the Monitoring Control Registry (MCR-ETL-02).

- Quarterly audit (Q3 scheduled: 2026-09-30) will reconcile alert definitions with SLOs and confirm that suppression windows map to approved CHGs.

Document Ownership and Versioning

- Steward: Director, Data Platform Operations.
- Control ID: CTRL-ETL-MON-0012; Version: 1.0 (Published 2026-07-15 15:15 UTC; CHG-0063199).
- Next Review: 2026-08-15 or following any material change to baselines or freshness objectives.

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Section 9: Alert Quality Review and Tuning Cycle (AQRTC)

Purpose and Scope

This section documents the standardized cadence and methodology used by Meridian Capital's Data Operations and ETL Reliability Engineering (ETLRE) teams to assess alert efficacy, reduce noise, and improve time-to-detect (TTD) and time-to-acknowledge (TTA) for ETL pipeline incidents. The Alert Quality Review encompasses platform-generated alerts (Enterprise Observability Platform, EOP), custom business KPI monitors, and data quality signals instrumented across ingestion, transformation, and publication layers. The intent is to ensure alerts are actionable, measured against defined SLOs, and aligned to current ownership models and runbook linkages.

Review Session Parameters

Field	Value
Review Window	2026-05-01 00:00 UTC to 2026-06-15 23:59 UTC
Facilitator	ETLRE (On-Call Rotation Lead: ROTA-ETLRE-04)
Participants	Data Ops L2, EOP Admin, Finance Analytics Rep, PMO Observer
Data Sources	EOP Alert Store v3.2, ServiceNow INC/PRB, JIRA MC-ETL board
Tickets Reviewed	117 alerts; 42 incidents (INC), 6 problems (PRB)
Reference Change Window	CHG-11992 (scheduler patch), CHG-12011 (connector uplift)
Meeting Timestamp	2026-06-17 09:00-10:30 UTC
Minutes and Artifacts	CONFL-ETL-2459 (AQRTC Minutes Q2-Wk3)

Alert Inventory Snapshot (Top-Impact Signals)

Alert ID	Signal Name	Severity	Volume	Ack Median	Runbook Ref	Owner
EOP-ETL-AL-0091	Job Duration P95 ..	P2	34	7m 12s	RBK-ETL-004-DUR	ETLRE
EOP-DQ-AL-0217	Record Count Vari..	P2	19	8m 30s	RBK-DQ-002-RCV..	Data Quality
EOP-SCH-AL-0033	Scheduler Missed ..	P1	5	4m 41s	RBK-SCH-001-WIN	Data Ops
EOP-FR-AL-0146	Data Freshness > ..	P2	27	9m 02s	RBK-ETL-006-FR..	ETLRE
EOP-SCHEMA-02..	Schema Drift (non-..	P3	12	14m 18s	RBK-DQ-005-SCH..	Data Quality

Meeting Log Excerpt (UTC)

- 2026-06-17 09:04:13 - Facilitator confirms quorum; acknowledges linkage to CHG-11992 impact window. Reference: INC-279114 and PRB-8821.
- 2026-06-17 09:17:56 - EOP Admin presents false positive analysis for EOP-ETL-AL-0091 (duration). Precision trend improved from 61% to 74% post scheduler patch; residual noise associated with end-of-quarter batch surge (FIN-EOQ tag).
- 2026-06-17 09:33:21 - Data Quality notes 3 instances of EOP-DQ-AL-0217 where expected source-side promotions altered baseline; proposes dynamic baselining with business calendar metadata (tag: CAL-FIN-HOL).
- 2026-06-17 09:48:02 - Data Ops highlights missed-window alert (EOP-SCH-AL-0033) correlation with maintenance holds; recommends enforcement of suppression during authorized holds (HOLD-MAINT tag) to reduce non-actionable P1s.
- 2026-06-17 10:06:47 - PMO requests formal metric for "Actionability Rate" (alerts leading to ticketed remediation

within 30 minutes).

- 2026-06-17 10:24:19 - Consensus to merge duplicate freshness and duration alerts for pipeline MC-ETL-PIPE-042 (redundant triggers verified). Action assigned to ETLRE.

Findings Summary

- Actionability Rate (Target >= 80%): Observed 76% across period; most gaps linked to seasonal baseline shifts and untagged maintenance holds.
- MTTA (Median Time to Acknowledge) for P1: 4m 41s (Target <= 5m) - within target; P2: 8m 51s (Target <= 8m) - marginally out of target during two surge days (2026-05-31, 2026-06-01).
- Alert Precision (true positive ratio) improved by 9 percentage points post CHG-11992; further uplift contingent on calendar-aware baselines.
- Runbook Coverage: 96% of active alerts linked to a current runbook reference; 4% require updates due to ownership changes post team restructure (REF: ORG-ETL-2026Q2).

Alert Tuning Backlog (Approved)

Backlog ID	Alert ID	Proposed Change	Rationale	Owner	Due Date	Ticket Ref
ATB-221	EOP-DQ-AL-0217	Enable calendar-a..	Reduce false positi..	DQ Team	2026-07-05	MC-ETL-4231
ATB-222	EOP-SCH-AL-0033	Auto-suppress on ..	Avoid non-actiona..	DataOps	2026-06-28	MC-ETL-4234
ATB-223	EOP-ETL-AL-0091	Merge with freshn..	Consolidate overla..	ETLRE	2026-07-02	MC-ETL-4238
ATB-224	EOP-SCHEMA-02..	Elevate severity on..	Improve responsiv..	DQ Lead	2026-07-09	MC-ETL-4240

Key Metrics (Baseline vs Target)

Metric	Baseline (Review Window)	Target (Q3)	Notes
Actionability Rate	76%	>= 85%	Introduce calendar baselines; r..
MTTA P1	4m 41s	<= 4m 30s	Minor optimization via routing t..
MTTA P2	8m 51s	<= 7m 30s	Reduce noise to boost attention
Alert Precision (All)	74%	>= 82%	Two-step filtering; de-dup on pi..
Runbook Linkage Coverage	96%	100%	Assign owners for stragglers
SLO Breach Notifications/Month	11	<= 6	Expected reduction after ATB-..

Procedural Audit Checks Completed

- Ownership and Routing: All P1/P2 alerts verified to reference current on-call alias groups: ONCALL-ETLRE, ONCALL-DQ, ONCALL-DATAOPS. Outdated alias ONCALL-ETL-LEGACY removed. Ticket: MC-ETL-4239.
- Metadata Hygiene: 97% of alerts contain asset tags (PIPE-ID, SYS-ENV, DATA-DOMAIN). Action ATB-225 opened for remaining 3% to add domain tags (RISK, FI-LEDGER, CLIENT-REF).
- Correlation Rules: Preliminary de-duplication rule (EOP-CORR-114) successfully reduces duplicate trigger pairs for duration/freshness by 38% on pilot pipelines PIPE-042, PIPE-091.

Risk Assessment and Next Steps

Residual operational risk remains moderate due to known seasonal volatility and incomplete alignment between business calendars and statistical baselines. Once ATB-221 and ATB-223 are implemented, expected risk will reduce to low. ETLRE to provide a mid-cycle checkpoint by 2026-07-12 with updated precision and MTTA metrics. PMO to track KPIs in Monthly Service Review (MSR) deck (REF: MSR-ETL-2026-07). No exceptions or deviations approved.

Approvals and Sign-Off

Role	Name/Group	Decision	Timestamp (UTC)
ETLRE Lead	ETLRE-LEAD-02	Approve	2026-06-17 10:28:33
Data Ops Manager	DATAOPS-MGR-01	Approve	2026-06-17 10:29:11
DQ Service Owner	DQ-OWNER-03	Approve	2026-06-17 10:29:54
PMO Representative	PMO-ETL-PORTFOLIO	Note	2026-06-17 10:30:00

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Section 10: KPI and SLO Monitoring Standards

Overview

This section enumerates the canonical KPI and SLO set for ETL pipeline monitoring at Meridian Capital. It defines metric semantics, calculation windows, target thresholds, and ownership such that dashboards, alerts, and service reviews present consistent and auditable views of operational health. All metrics are sourced from the Enterprise Observability Platform (EOP) and reconciled against the ETL Job Registry (EJR) to ensure complete coverage.

Metric Catalog (Operational and Data Quality)

Metric Name	Definition (Summ..	Window	Target/Threshold	Source	Owner		
Pipeline Success ..	Successful runs / t..	Rolling 30 days	>= 99.0%	EOP	Data Ops		
Data Freshness (m..	Lag between sche..	Daily P95	<= 30 min P95	EOP	ETLRE		
Job Duration P95	95th percentile of j..	Weekly	<= baseline + 15%	EOP	ETLRE		
Record Count Vari..		actual - expected	/ expected	Daily	<= 10% (standard)..	EOP	Data Quality
Schema Drift Freq..	Count of non-brea..	Monthly	<= 2 per domain	EOP	Data Quality		
SLO Breach Rate	Breaches / opportu..	Monthly	<= 2%	EOP	Service Owner		
Alert Actionability	Alerts leading to re..	Monthly	>= 85%	EOP	ETLRE		

Service Level Objectives (SLO) Definitions

Service Component	SLO Statement	Target	Measurement Notes
Ingestion Availability	Ingestion completes within win..	99.5% monthly	Window per EJR schedule; ex..
Publish Timeliness	Data visible in consuming layer..	98.0% monthly	Timeliness based on freshness..
Data Accuracy	Record counts within variance ..	99.0% monthly	Domain-specific thresholds ho..
Stability	P1 incident volume	<= 3 per month	Excludes declared maintenance

Monthly Service Review (MSR) Agenda and Artifacts

The MSR consolidates KPI/SLO performance, alert tuning outcomes, and backlog status. The following structure supports consistent coverage across teams while enabling traceability to remediation tickets.

MSR Session Record (Example)

Field	Value
Session Date	2026-07-08 14:00-15:00 UTC
Facilitator	Service Owner - ETL Platform
Attendees	ETLRE, Data Ops, Data Quality, EOP Admin, Finance Analyst
Artifacts	MSR Deck (MSR-ETL-2026-07), KPI Export (KPI-EXP-0708)
Decisions Logged	DEC-ETL-2026-07-01 to -05

Agenda (Timestamped Excerpt)

- 14:03 - Review of June KPI deltas vs May baseline; confirm improvement in Actionability to 82%.
- 14:15 - SLO Breach analysis: Two timeliness breaches traced to upstream source slowdown (tag: SRC-LATENCY). Confirmed no internal capacity constraints.

- 14:28 - Data Quality spotlight: GL domain variance at 6.2% on 2026-06-29; correction approved to tighten GL threshold to 5% with calendar-aware adjustment.
- 14:41 - Backlog review: ATB-221, ATB-223 on target; ATB-224 deferred by 1 week due to resource contention (note: no risk elevation).
- 14:53 - Action register updates; owners confirm closure timelines and dashboard annotations.

Data Quality Threshold Matrix (Domain-Specific)

Data Domain	Variance Threshold	Freshness SLO (P95)	Exceptions Policy Ref..	Owner
General Ledger	<= 5%	<= 20 min	DQ-EX-GL-2026	DQ-GL Lead
Client Master	<= 10%	<= 30 min	DQ-EX-CM-2026	DQ-CM Lead
Market Data	<= 8%	<= 15 min	DQ-EX-MD-2026	DQ-MD Lead
Risk Factors	<= 10%	<= 30 min	DQ-EX-RF-2026	DQ-RF Lead

Dashboard Standards and Field Mapping

All KPIs are surfaced in the ETL Service Health dashboard. Field names and tags must map consistently to the ETL Job Registry keys to ensure reliable filtering and roll-ups.

Dashboard Mapping Parameters

Dashboard Field	Source Field (EOP)	Registry Key (EJR)	Required Tags
Pipeline ID	pipeline_id	EJR.PIPE_ID	SYS-ENV, DATA-DOMAIN
Schedule Window	schedule_window	EJR.WIN_ID	CAL-FIN, HOLD-MAINT
Duration P95	job_duration_p95	EJR.BASELINE_DUR	FIN-EOQ (as applicable)
Freshness P95	freshness_p95	EJR.BASELINE_FRESH	SRC-SLA
Record Variance	count_variance_percent	EJR.BASELINE_RC	DOMAIN-THRESH

Operational Validation Checklist (Monthly)

- Confirm metric definitions unchanged in EOP release notes; no semantic drift observed (Ref: EOP-RN-3.2.4).
- Validate that 100% of P1/P2 pipelines carry required tags for calendar and maintenance holds; sample checks across three domains.
- Reconcile EOP pipeline inventory with EJR registry count; discrepancies documented (target: <= 1% variance).
- Review top 10 alert contributors to SLO breaches; ensure linkage to active backlog items.
- Ensure MSR decisions have corresponding tickets and due dates captured in JIRA (MC-ETL board).

Alert Routing Matrix (Excerpt)

Category	Severity Map	Primary Team	On-Call Alias	Escalation Window
Scheduler Missed Wind..	P1 (critical)	Data Ops	ONCALL-DATAOPS	10 min
Freshness Breach	P2 (high)	ETLRE	ONCALL-ETLRE	15 min
Count Variance	P2 (high)	Data Quality	ONCALL-DQ	20 min
Schema Drift	P3 (moderate)	Data Quality	ONCALL-DQ	30 min
Duration Anomaly	P2 (high)	ETLRE	ONCALL-ETLRE	20 min

Change Control and Traceability

All modifications to KPI definitions, SLO targets, or routing rules require formal change control with clear audit trails.

The following items were processed in the current cycle:

- CHG-12011: Connector uplift--validated no adverse impact to duration calculations (post-change observation period: 7 days; no anomalies recorded).
- CHG-12032: Dashboard field remap--aligned freshness calculation to exclude authorized maintenance holds; verified against sample set PIPE-042, PIPE-075, PIPE-091.

Exceptions and Non-Conformances

No exceptions to standards were requested or approved in the period. One non-conformance identified during inventory reconciliation (two pipelines missing DATA-DOMAIN tags) tracked under MC-ETL-4245 with due date 2026-06-30. No client-impacting risk identified; monitoring coverage unaffected.

Reporting and Communication

Weekly summary of KPI/SLO performance is distributed to the ETL leadership distribution list with links to the live dashboard and the MSR action tracker. Where targets are not met, the responsible owner includes remediation status, expected recovery timeline, and cross-team dependencies. All communications reference ticket identifiers and maintain consistent timestamping in UTC to support audit requirements.

End of Page 14.

Page 15 -- Section 10: Alert Hygiene Program Outcomes (FY26 Q2-Q3)

Overview

This section documents the results of Meridian Capital's Alert Hygiene Program executed between 2026-05-01 and 2026-07-15 for the ETL Core and Satellite Pipelines portfolio. Objectives were to reduce non-actionable notifications, tighten routing accuracy, and validate escalation paths without impacting Mean Time to Detect (MTTD) or breach rates for business Service Level Objectives (SLOs). Program sponsorship: Enterprise Monitoring Office (EMO). Execution owners: Data Platform Operations (DPO), Risk Technology ETL (RTech-ETL), and Analytics Engineering (AE).

Change and Governance References

- Change Requests: CRQ-2026-0630-0192 (threshold rationalization), CRQ-2026-0705-0276 (routing updates), CRQ-2026-0708-0113 (suppression window catalog)
- Jira Epics: MCETL-4816 (Alert Inventory), MCETL-4821 (Noise Reduction), MCETL-4859 (Routing QA)
- Effective Dates: initial rollout 2026-07-09 01:00 UTC; monitoring freeze window 2026-07-09 01:00-05:00 UTC; stabilization review 2026-07-12 14:30 UTC
- Stakeholders: DPO On-Call (rotations DPO-OC-23A, DPO-OC-23B), RTech-ETL (rtech-etl-ops), EMO (emo-governance)

Alert Policy Parameters (Before vs After)

| Policy ID | Monitored Entity | Prior Threshold | New Threshold | Evaluation Window | Auto-Suppress Rule | Owner |

| MC-AL-ETL-021 | Daily Batch Duration (Core Batch A) | > 120 min | > 135 min | 3 consecutive runs | Suppress on "Bank Holiday Calendar = True" | DPO |

| MC-AL-ETL-044 | Late File Arrival (MarketData FX) | Delay > 10 min post-SLA | Delay > 12 min post-SLA | Rolling 30 min | Suppress during vendor advisory flag | RTech-ETL |

| MC-AL-ETL-067 | Orchestrator Queue Depth | > 250 queued tasks | > 300 queued tasks | 5 min average | None | DPO |

| MC-AL-ETL-089 | Source API Error Rate (non-critical endpoints) | > 1.5% | > 2.0% | 10 min | Suppress during planned maintenance windows | AE |

| MC-AL-ETL-102 | Partition Skew Alert (Transform Set C) | 3x median | 3.5x median | 2 jobs | None | AE |

Observations and Findings

- Post-change alert volume reduced by 28.4% (baseline: 1,182 alerts/week; post: 846 alerts/week; sample window: 2026-07-09 to 2026-07-15).
- False-positive rate (alerts closed "No Action Required") decreased from 41% to 17%.
- MTTD unchanged at 2.8 minutes (median) for high-severity signals; MTTA improved from 9.4 minutes to 6.7 minutes due to clearer routing and deduplication.
- No SLO breaches attributable to the updated thresholds or suppression rules in the observation window; compliance with batch SLO (Core Batch A complete by 03:30 UTC) maintained at 99.2%.

Correlated Ticket Summary (Representative)

- MC-INC-2026-07-11-0421 (P2): Late File Arrival (MarketData FX) occurred at 2026-07-11T04:21:33Z; resolved at 2026-07-11T04:38:11Z. Root cause: vendor advisory flag active; suppression correctly applied for informational alerts; single actionable alert routed to RTech-ETL.
- MC-INC-2026-07-12-0157 (P3): Queue depth transient; 7-minute spike during peak scheduling; self-corrected; alert volume decreased from 6 to 1 due to evaluation window change.

- MC-PBI-2026-07-13-1032 (Problem): Partition skew intermittent; ongoing review under MCETL-4932; no user impact.

Monthly Alert Rule Certification Procedure (Non-Technical Artifacts)

| Step | Activity | Responsible Team | Required Inputs | Outputs | System of Record |

| 1 | Export active alert catalog (metadata only) | EMO | Monitoring registry snapshot | Certified inventory (.csv) | EMO-CERT-REG |

| 2 | Review thresholds against last 90 days of metrics | DPO, AE | Metric summaries; run statistics | Signed threshold verification memo | SharePoint-ETL-Controls |

| 3 | Validate routing and on-call mappings | DPO | On-call roster; escalation matrix | Updated routing table | Routing-Index v2 |

| 4 | Negative test using historical replay (no live system changes) | EMO | Sanitized alert events | Pass/fail report | QA-Lab-Alerts |

| 5 | Approvals and change scheduling | EMO, RTech-ETL | CRQ drafts | Approved CRQ entries | Change Calendar |

Known Benign Conditions Catalog (Excerpt)

| Condition ID | Description | Suppression Window | Justification | Review Ticket |

| KBC-ETL-005 | Vendor "FX MarketData" delay on Mondays (02:55-03:05 UTC) | 10 min weekly (Mon) | Vendor-side file preparation cycle; no downstream SLA impact | MCETL-4762 |

| KBC-ETL-011 | Bank Holiday calendar alignment drift (US/UK) | Holiday dates only | Source feed timestamp variance; batch completion SLO unaffected | MCETL-4816 |

| KBC-ETL-019 | Orchestrator warm-up post-maintenance | First run after maintenance | Expected queue priming; stabilized within 6 minutes | MCETL-4851 |

KPI Posture (Alerting)

- Alert Volume (weekly): Target <= 900; Actual 846; Status: On Track
- MTTA (P1/P2): Target <= 8 min; Actual 6.7 min; Status: Improved
- Deduplication Efficiency: Target >= 30%; Actual 33%; Status: Met
- Routing Accuracy: Target >= 95%; Actual 97.2%; Status: Exceeds

Audit Notes

- 2026-07-08T13:44:10Z EMO confirmed readiness for rollout; CRQ-2026-0630-0192 approved.
- 2026-07-09T05:03:22Z DPO validated no missed P1 signals in first four hours post-change.
- 2026-07-12T15:02:55Z Stabilization review closed with no corrective actions; next cycle scheduled 2026-08-05.

Risks and Mitigations

- Risk: Over-suppression during unforeseen vendor anomalies. Mitigation: Suppression windows capped; EMO requires monthly certification; incident overrides remain unaffected.
- Risk: Threshold drift relative to seasonal volume. Mitigation: Seasonal review cadence introduced (Nov/Feb/May/Aug) with retrospective baselining.

Action Items

- AE to complete MCETL-4932 (Partition Skew Tuning) by 2026-08-02.
- DPO to extend routing QA to Satellite Pipelines (MCETL-4970) by 2026-08-09.

Page 16 -- Section 11: Capacity and Performance Observability

Overview

This section provides the current capacity and performance observability posture for Meridian Capital's ETL platform, focusing on batch window utilization, backlog behavior, and data freshness SLO adherence. Data period: 2026-06-15 through 2026-07-15. Systems observed include Orchestrator (prod cluster), Transform Engines (Spark-based compute pools), and Storage (curated zones for Risk, Finance, and MarketData domains). Owners: DPO (capacity), AE (performance baselines), EMO (SLO governance).

Performance Baseline Update (Q3)

- Core Batch A average duration: 102 min (p50), 118 min (p95) versus window of 150 min. Headroom: 32 min (p95).
- Transform Engine utilization (peak window 01:30-03:30 UTC): 62% (p50), 74% (p95), max observed 81%. No sustained saturation events recorded.
- Input data variance: Late arrivals (>10 min post-SLA) at 1.2% of feeds; no clustering outside known benign windows (see KBC-ETL-005).

Metric Dictionary (Excerpt)

| Metric | Definition | Unit | Source | Retention |

| batch_duration_minutes | End-to-end runtime for named batch | Minutes | Orchestrator telemetry | 400 days |

| backlog_queue_depth | Pending tasks awaiting executor slots | Count | Orchestrator telemetry | 400 days |

| executor_utilization_pct | Active executor time / total available | Percent | Compute pool telemetry | 400 days |

| data_freshness_offset | Time delta between expected and actual dataset availability | Minutes | Dataset registry | 400 days |

| sla_breach_count | Count of completed runs exceeding SLO | Count | EMO SLO engine | 400 days |

| late_file_arrival_rate | Percent of files arriving after planned SLA | Percent | Source ingestion logs | 400 days |

SLA/SLO Scorecard (Representative)

- Core Batch A completion by 03:30 UTC: 99.2% compliance (Target >= 99.0%)
- Finance Aggregation refresh by 04:15 UTC: 99.6% compliance (Target >= 99.0%)
- MarketData daily ingest availability by 02:45 UTC: 98.8% compliance (Target >= 98.5%)
- Platform availability (orchestrator, prod): 99.97% (rolling 30 days)

Backlog Dynamics Summary

During the reporting period, the 95th percentile backlog_queue_depth remained under 210, with intermittent peaks associated with coordinated upstream feed arrivals. Queue aging analysis indicates a median task wait of 1.9 minutes, improving to 1.4 minutes post-threshold rationalization (see Section 10). No evidence of prolonged queue contention; auto-scaling events aligned with expected arrival patterns as per change record CRQ-2026-0705-0276 (routing updates did not materially alter scheduling cadence).

Batch Window Risk Commentary

- Risk Hotspots: 02:20-02:40 UTC shows the narrowest margin for Core Batch A due to simultaneous ingestion bursts. Even under p95 durations, the residual headroom remained >= 18 minutes.
- Sensitivity: Model simulations (historical replay; no live impact) suggest that a 10% increase in input volume

would elevate p95 runtime to 125-128 minutes, preserving SLO margin.

- Recommendation: Maintain current executor allocation baseline; recheck after the month-end close period due to predictable volume spikes.

Backlog Escalation Workflow (Non-Invasive)

| Step | Trigger Condition | Responsible Team | Inputs | Expected Outcome | Ticket Reference |

| 1 | backlog_queue_depth > 300 for > 10 min | DPO On-Call | Orchestrator dashboard snapshot | Confirm sustained condition | MC-ESC-ETL-2026-07-15-001 |

| 2 | executor_utilization_pct >= 85% (p90, 15 min) | DPO + AE | Utilization report | Assess compute headroom | Linked to MC-ESC-...-001 |

| 3 | data_freshness_offset forecast > 12 min on any SLO dataset | EMO | SLO predictor | Notify impacted app owners | MC-NFY-2026-07-15-014 |

| 4 | If condition normalizes within 20 min | DPO | Time series review | Document and close escalation | Closure in MC-ESC-...-001 |

| 5 | If condition persists past 20 min | DPO + Stakeholders | Stakeholder list | Initiate standard incident P3 | Create MC-INC per protocol |

Data Freshness SLO Tracking

Data freshness offsets for curated datasets in Risk and Finance domains remained within tolerance. The largest observed offset was 9 minutes (MarketData FX on 2026-07-11), associated with a vendor advisory flag. EMO verified that suppression of non-actionable notifications did not conceal any breach conditions; all SLO-critical alerts retained immediate routing. The SLO predictor (using historical run metadata) correctly flagged two potential minor offsets; both normalized without intervention.

Capacity Outlook

- Forecast: Based on trailing 90-day growth (2.3% weekly increase in task count), the platform is projected to reach 75-78% p95 executor utilization by mid-quarter. The existing buffer is considered adequate.
- Contingency: Maintain review cadence with AE to refresh baselines post month-end. No changes recommended to the execution pool sizing before 2026-08-15 barring unexpected upstream variability.

Quality Checks and Controls

- 2026-07-10T12:06:41Z: EMO audit confirms SLO compliance calculation aligned to UTC with dataset-specific grace windows documented in SLO registry.
- 2026-07-13T03:48:09Z: DPO verification of backlog measurement labels; no discrepancies between orchestrator telemetry and EMO rollups.

Open Items

- MCETL-4978: Evaluate granularity of executor_utilization_pct at job-class level for Satellite Pipelines; due 2026-08-07.
- MCETL-4983: Extend data_freshness_offset measurement to include confidence intervals; due 2026-08-12.

Notes

All times reported in UTC. No changes to production configuration were made as part of this reporting section; analyses relied on telemetry and historical replay within controlled environments.

Page 17 -- Appendix A: KPI Catalog, Sampling Windows, and Alert Review Log (Production)

This appendix consolidates production KPI definitions, current sampling windows, and the most recent alert review observations for Meridian Capital's enterprise ETL estate. The objective is to provide a stable reference for on-call rotation stakeholders (Data Platform Operations, Infra Observability, Risk Analytics ETL, and Enterprise Data Governance) without prescribing implementation instructions. This is an attestation of baselines, thresholds, ownership, and review cadence as of Q3 FY2026.

KPI Catalog (Production)

| Metric ID | Metric Name | Pipeline Scope | Source System | Sampling Window | Aggregation | Alert Threshold | SLO Target | Owner (Group) | Last Review (UTC) | Ticket Ref |

| KPI-001 | Throughput Rows/Min | ETL-ACCT-DAILY-001 | CoreLedger | 5 min sliding | Mean | Warn < 120k; Crit < 80k | >= 99.5% daily volume | Data Platform Ops | 2026-07-08 14:30:00Z | MC-CAP-REV-2026-071 |

| KPI-002 | End-to-End Latency | DAG-FIN-SETTLE-P1 | PaymentsBus | 15 min discrete | 95th pct | Warn > 42m; Crit > 55m | 95th <= 40m | Infra Observability | 2026-07-09 11:05:14Z | MC-OBS-REV-2026-122 |

| KPI-003 | File Arrival Variance | ETL-TX-INGEST-002 | SFTP Gateway | 10 min discrete | Std Dev | Warn > 2.5sigma | <= 2sigma | Data Platform Ops | 2026-07-10 07:50:32Z | MC-DPO-REV-2026-309 |

| KPI-004 | Transformation Error Rate | DAG-RISK-CALC-B2 | RiskCalcSvc | 15 min sliding | Error% | Warn >= 0.30%; Crit >= 0.50% | < 0.25% | Risk Analytics ETL | 2026-07-12 19:40:06Z | MC-RSK-REV-2026-044 |

| KPI-005 | SLA Breach Count (Daily) | Portfolio Snapshots | EDW | 24h window | Count | Warn >= 2; Crit >= 4 | 0 per day | Data Governance | 2026-07-11 13:15:20Z | MC-DG-REV-2026-218 |

| KPI-006 | Queue Depth (Slots) | ETL Schedulers | Orchestrator | 1 min instant | Max | Warn >= 75%; Crit >= 90% | Peak <= 70% | Infra Observability | 2026-07-13 05:28:47Z | MC-OBS-REV-2026-126 |

| KPI-007 | Upstream Freshness | Market Data Cache | MarketFeed | 10 min discrete | Age (min) | Warn >= 18; Crit >= 30 | <= 15 min | Risk Analytics ETL | 2026-07-13 22:14:59Z | MC-RSK-REV-2026-047 |

| KPI-008 | Successful Loads | ETL-ACCT-DAILY-001 | CoreLedger | 24h window | Count | Warn < 3; Crit < 2 | >= 3 per day | Data Platform Ops | 2026-07-08 14:30:00Z | MC-CAP-REV-2026-071 |

KPI Review Summary (Q3 FY2026)

- Review window: 2026-07-08 to 2026-07-13; convened by Data Platform Operations with Infra Observability and Risk Analytics ETL.
- Key outcome: KPI-002 latency thresholds raised by +2 minutes to align with month-end payroll batch cohabitation; no SLA impact recorded in prior 90 days (Ref: MC-OBS-ANL-2026-310).
- KPI-004 error rate alert routing retained to Risk Analytics ETL primary on-call; persistent anomalies beyond 45 minutes to be co-notified to Data Platform Ops per existing escalation matrix (no changes to policy).
- KPI-006 queue depth remained within historical norms. Secondary threshold retained for weekend maintenance windows to avoid false positives during scheduled resource reservation (Ref: CHG-2026-07-PRD-1420, approved).

Observation Log Excerpts (Q2-Q3 FY2026, Production)

| Timestamp (UTC) | Event ID | Severity | Component | Finding | Follow-Up Ticket | Status |

| 2026-07-14 02:10:45Z | EVT-062713 | Warning | DAG-FIN-SETTLE-P1 | 95th latency peaked at 44m for 20m | MC-SNOW-INC-2026-004182 | Resolved 2026-07-14 03:02Z |

| 2026-07-05 23:58:02Z | EVT-062482 | Info | ETL-ACCT-DAILY-001 | Throughput dipped 11% vs 7-day mean | MC-JIRA-ANL-23145 | Closed--no action |

| 2026-06-30 04:21:19Z | EVT-062219 | Warning | Schedulers (PRD) | Queue depth sustained 78% for 12m | MC-SNOW-INC-2026-003991 | Resolved 2026-06-30 04:49Z |

| 2026-06-18 06:44:53Z | EVT-061853 | Warning | DAG-RISK-CALC-B2 | Error rate 0.36% over 10m window | MC-RSK-PRB-2026-0018 | Problem queued |

| 2026-05-27 01:33:07Z | EVT-061325 | Info | Market Data Cache | Freshness exceeded 16m (below warn) | N/A | Monitored |

Narrative Findings

- The July 14 latency excursion aligned with a third-party upstream batch overlap. No SLA breach; MTTR was 52 minutes with automated dampening preventing duplicate alerts during the interval (Ref: MC-OBS-ANL-2026-327).
- Queue depth on June 30 reflected pre-approved resource reservation ahead of a platform patch window. Alert was informational and closed after validation against the change calendar.
- No material deviation in daily volume attainment for accounting loads during the quarter; KPI-001 tracked within 1.2 standard deviations of the trailing 30-day mean.

Parameter Snapshot: Production Orchestrators and Windows

| Environment | Scheduler ID | Window Start | Window End | Max Concurrency | Backpressure Policy | Last Change Ref | Reviewer |

| PRD | sch-prd-eus2-01 | 00:00Z | 06:00Z | 280 slots | Defer non-critical queues >= 75% | CHG-2026-07-PRD-1420 | Infra Obs |

| PRD | sch-prd-eus2-02 | 06:00Z | 12:00Z | 260 slots | Hold bursty queues at warn | CHG-2026-06-PRD-1188 | Data Platform Ops |

| PRD | sch-prd-eus2-03 | 12:00Z | 18:00Z | 300 slots | Smoothing for top-10 DAGs | CHG-2026-05-PRD-1003 | Data Platform Ops |

| PRD | sch-prd-eus2-04 | 18:00Z | 24:00Z | 240 slots | Defer non-critical queues >= 75% | CHG-2026-04-PRD-0914 | Infra Obs |

Governance Notes

- Data Governance validation for KPI naming and lineage references completed 2026-07-11 (MC-DG-REV-2026-218). No taxonomy changes recommended.
- All KPIs are mapped to service catalogs FIN-ETL-SVC-001 through FIN-ETL-SVC-004 with owner attestation current as of 2026-07-13.
- No exceptions requested or granted in the review period. All deviations logged as informational during approved change windows.

Upcoming Review Plan

- Monthly KPI health review: 2026-08-12 14:00-14:45Z, chaired by Data Platform Ops (invite: MC-CAL-2026-ETL-MON-08).
- Quarter-end threshold calibration checkpoint: 2026-09-27 10:00-10:30Z, cross-functional (Infra Observability, Risk Analytics ETL, Data Governance).

Distribution

- Primary: Data Platform Operations; Infra Observability; Risk Analytics ETL; Enterprise Data Governance.
- Secondary: Business Operations (Portfolio Accounting) for situational awareness.

Documented by: S. Hsu (Data Platform Ops), 2026-07-14 15:22:11Z. Peer-reviewed by A. Petrov (Infra Observability), 2026-07-14 16:05:44Z.

Page 18 -- Appendix B: Runbook QA Controls, Change Calendar, and Attestations

This appendix provides quality assurance evidence for ETL monitoring runbooks, a rolling 30-day change calendar excerpt for production schedulers and dependent assets, and closing attestations. The materials below reflect operational reality as of 2026-07-15 and are intended for audit consumption and stakeholder alignment.

Runbook QA Checklist (Evidence Captured)

| Step No. | Control ID | Control Name | Evidence Artifact | Owner | Timestamp (UTC) | Result |

| 1 | MON-CTRL-001 | KPI Definition Currency | MC-Repo-ETL-MON/v5.12 tag | Data Platform Ops | 2026-07-10 09:12:28Z | Pass |

| 2 | MON-CTRL-004 | Alert Route Verification | Pager roster export 2026-07-09 | Infra Observability | 2026-07-09 18:03:10Z | Pass |

| 3 | MON-CTRL-006 | Dependency Mapping Completeness | Lineage snapshot LN-2026-07-08 | Data Governance | 2026-07-08 13:41:05Z | Pass |

| 4 | MON-CTRL-009 | Sampling Window Audited | Prom baselines rpt 2026-07-08 | Infra Observability | 2026-07-08 15:55:22Z | Pass |

| 5 | MON-CTRL-011 | Runbook Cross-Links Valid | Link check rpt LC-2026-07-07 | Data Platform Ops | 2026-07-07 20:27:33Z | Pass |

| 6 | MON-CTRL-015 | Escalation Matrix Current | On-call matrix OCM-2026-Q3 | Data Platform Ops | 2026-07-06 10:18:49Z | Pass |

| 7 | MON-CTRL-018 | Post-Incident Template Up to Date | PIR template v3.4 | Risk Analytics ETL | 2026-07-05 08:06:11Z | Pass |

| 8 | MON-CTRL-020 | Change Calendar Synced | CAB log 2026-W27 | Change Mgmt Office | 2026-07-04 12:19:07Z | Pass |

RACI Snapshot for Monitoring and Alerting (P1-P3, Production)

| Activity | Data Platform Ops | Infra Observability | Risk Analytics ETL | Business Ops | Data Governance |

| KPI threshold review | A/R | C | C | I | C |

| Alert route maintenance | R | A/R | C | I | I |

| Incident triage (P1) | A/R | R | C | I | I |

| Incident triage (P2-P3) | R | R | C | I | I |

| Post-incident review | A | C | R | C | C |

| Metric catalog maintenance | R | C | C | I | A |

| Change communication | C | C | C | A/R | C |

Change Calendar Excerpt (Rolling 30 Days, Production)

| Change ID | Window (UTC) | Scope | Impact Class | Risk | Approvals | Backout Plan ID | Status |

| CHG-2026-07-PRD-1420 | 2026-07-13 02:00-03:00 | sch-prd-eus2-01 param tune | Low | Low | CAB-2026-W28, OpsMgr | BOP-2026-07-01 | Completed |

| CHG-2026-07-PRD-1467 | 2026-07-20 01:30-03:30 | DAG-FIN-SETTLE-P1 window realign | Medium | Med | CAB-2026-W29, AppOwner | BOP-2026-07-03 | Scheduled |

| CHG-2026-07-PRD-1491 | 2026-07-23 04:00-05:00 | KPI catalog annotation update | Low | Low | CAB-2026-W30 | BOP-2026-07-05 | Scheduled |

| CHG-2026-08-PRD-1505 | 2026-08-02 02:00-04:00 | Orchestrator rotation test (non-peak) | Medium | Med | CAB-2026-W31, SRE | BOP-2026-08-01 | Draft |

| CHG-2026-08-PRD-1512 | 2026-08-05 23:00-23:45 | Market data freshness tag alignment | Low | Low | CAB-2026-W31, Governance | BOP-2026-08-02 | Draft |

Metrics Trend Summary (Q2 FY2026 to Q3 MTD, Production)

| Month | Total Rows Processed (Bn) | Success Rate | Mean E2E Latency | P95 E2E Latency | Alert Count | MTTA |

| 2026-04 | 12.4 | 99.93% | 18m 42s | 34m 10s | 118 | 7m 12s |

| 2026-05 | 13.1 | 99.95% | 18m 03s | 33m 55s | 104 | 6m 58s |

| 2026-06 | 13.6 | 99.92% | 19m 11s | 35m 02s | 127 | 7m 21s |

| 2026-07 (MTD) | 6.2 | 99.94% | 18m 57s | 34m 28s | 52 | 6m 49s |

Interpretation

- Alert volume remains within seasonal norms; month-end boundaries and payroll cycles continue to drive predictable P95 increases without SLA breaches.
- MTTA improvements correspond to roster stabilization since 2026-05 and consistent paging hygiene along existing escalation paths.
- Success rate variance in June traces to a single-day partial reprocessing in RiskCalcSvc (Ref: MC-RSK-PRB-2026-0016), fully reconciled within accounting cutoffs.

Glossary of Monitored Identifiers

- ETL-ACCT-DAILY-001: Daily Accounting Load (CoreLedger to EDW).
- DAG-FIN-SETTLE-P1: Settlement DAG, Priority 1, finance pipelines.
- DAG-RISK-CALC-B2: Batch risk calculations for book B2.
- sch-prd-eus2-0X: Production orchestrator schedulers for East US region.
- KPI-00X: Monitoring KPIs aligned to FIN-ETL-SVC service catalog entries.
- Market Data Cache: Normalized intraday price cache feeding RiskCalcSvc.
- MC-SNOW-INC-YYYY-XXXXXX: Incident tracking references in ServiceNow.
- CHG-YYYY-MM-PRD-XXXX: Production change control references via CAB.

Runbook QA Notes and Evidence Index

- Lineage snapshots LN-2026-07-08 and LN-2026-06-10 archived under EDG-ARCHIVE-2026-Q3; cross-checked against active DAG manifests on 2026-07-08 with no orphaned nodes detected.
- KPI ownership matrix validated against HR roster effective 2026-07-01; temporary assignment coverage documented for summer leave periods (Ref: DPO-ROTA-2026-SUMMER).
- PIR templates v3.4 incorporate standard sections for observed vs. expected baselines, without modifying escalation or approval flows.

Attestations and Sign-offs

- Data Platform Operations: "KPI and alerting baselines reflect current load patterns and approved change windows." Signed: L. Ramirez (Mgr, Data Platform Ops), 2026-07-15 12:14:02Z.
- Infra Observability: "Sampling windows and notification routes tested and verified; no outstanding defects." Signed: A. Petrov (Lead, Infra Observability), 2026-07-15 12:18:37Z.
- Risk Analytics ETL: "Risk calculation monitoring thresholds adequate for volatility observed in Q2; no exceptions requested." Signed: M. Okoye (Dir, Risk Analytics ETL), 2026-07-15 12:22:09Z.
- Enterprise Data Governance: "KPI taxonomy and lineage annotations current; controls MON-CTRL-006 and MON-CTRL-020 remain effective." Signed: R. Chen (Head, Data Governance), 2026-07-15 12:29:44Z.

Document Control

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- Supersedes: v5.11 (2026-04-10)

Closing Statement

This handbook's monitoring and alerting parameters, ownership attributions, and audit evidence are current as of the dates indicated. The materials herein are intended to streamline operational awareness and facilitate routine KPI health validation across production ETL services. Any changes to thresholds, sampling windows, or ownership will follow the standard change process with CAB visibility and will be reflected in the subsequent revision cycle. End of document.